Physica bk2

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1 On GlossySnake

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2 Physica bk2

PHYSICA Bk.2

PHYSICS Book 2

Κεφάλαιο 1

Chapter 1

Τῶν ὄντων τὰ μέν ἐστι φύσει, τὰ δὲ δι' ἄλλας αἰτίας,

Of the beings the on the one hand is by nature, the but through other causes,

φύσει μὲν τά τε ζῷα καὶ τὰ μέρη αὐτῶν καὶ τὰ φυτὰ καὶ τὰ

by nature on one hand the and animals and the parts of them and the plants and the

ἁπλᾶ τῶν σωμάτων, οἷον γῆ καὶ πῦρ καὶ ἀὴρ καὶ ὕδωρ (ταῦτα

simple of the bodies, such as earth and fire and air and water (these

γὰρ εἶναι καὶ τὰ τοιαῦτα φύσει φαμέν), πάντα δὲ ταῦτα φαίνεται

for are also the such things by nature we say), all but these seem

διαφέροντα πρὸς τὰ μὴ φύσει συνεστῶτα. τούτων μὲν

different in relation to the not by nature conjoined. Of these on one hand

γὰρ ἕκαστον ἐν ἑαυτῷ ἀρχὴν ἔχει κινήσεως καὶ στάσεως, τὰ

for each in itself a principle has of motion and of rest, the

μὲν κατὰ τόπον, τὰ δὲ κατ' αὔξησιν καὶ φθίσιν, τὰ δὲ

on one hand according to place, the but according to increase and decrease, the but

κατ' ἀλλοίωσιν· κλίνη δὲ καὶ ἱμάτιον, καὶ εἴ τι τοιοῦτον

according to alteration; bed but and garment, and if anything such

ἄλλο γένος ἐστίν, ᾗ μὲν τετύχηκε τῆς κατηγορίας ἑκάστης

other kind is, in which indeed it has happened of the category each

καὶ καθ' ὅσον ἐστὶν ἀπὸ τέχνης, οὐδεμίαν ὁρμὴν ἔχει

and according to as far as it is from art, no impulse has

μεταβολῆς ἔμφυτον, ᾗ δὲ συμβέβηκεν αὐτοῖς εἶναι λιθίνοις

of change inherent, in which but it has happened to them to be stony

ἢ γηΐνοις ἢ μικτοῖς ἐκ τούτων, ἔχει, καὶ κατὰ τοσοῦτον, ὡς

or earthy or mixed from these, it has, and according to so much, as

οὔσης τῆς φύσεως ἀρχῆς τινὸς καὶ αἰτίας τοῦ κινεῖσθαι καὶ

there being of the nature principle of some and cause of the to be moved and

ἠρεμεῖν ἐν ᾧ ὑπάρχει πρώτως καθ' αὐτὸ καὶ μὴ κατὰ

to be at rest in which exists primarily according to itself and not according to

συμβεβηκός

accidental

(λέγω δὲ τὸ μὴ κατὰ συμβεβηκός, ὅτι γένοιτ' ἂν αὐτὸς

I say but the not according to what has happened, that it may be if he himself

αὑτῷ τις αἴτιος ὑγιείας ὢν ἰατρός· ἀλλ' ὅμως οὐ καθὸ

to himself someone responsible for health being a doctor. but however not insofar as

ὑγιάζεται τὴν ἰατρικὴν ἔχει, ἀλλὰ συμβέβηκεν τὸν αὐτὸν ἰατρὸν

he is healthy the medical has, but it has happened that same doctor

εἶναι καὶ ὑγιαζόμενον· διὸ καὶ χωρίζεταί ποτ' ἀπ'

to be and healthy. Therefore and he is distinguished at times from

ἀλλήλων). ὁμοίως δὲ καὶ τῶν ἄλλων ἕκαστον τῶν ποιουμένων·

one another). Similarly but also of the others each of the makers.

οὐδὲν γὰρ αὐτῶν ἔχει τὴν ἀρχὴν ἐν ἑαυτῷ τῆς ποιήσεως, ἀλλὰ

Nothing for of them has the origin in themselves of creation, but

τὰ μὲν ἐν ἄλλοις καὶ ἔξωθεν, οἷον οἰκία καὶ τῶν ἄλλων τῶν

the some in others and from outside, such as a house and of the others of the

χειροκμήτων ἕκαστον, τὰ δ' ἐν αὑτοῖς μὲν ἀλλ' οὐ καθ'

hand-formed each, the and in themselves indeed but not according to

αὑτά, ὅσα κατὰ συμβεβηκὸς αἴτια γένοιτ' ἂν αὐτοῖς.

themselves, as much according to contingent cause might be if to themselves.

φύσις μὲν οὖν ἐστὶ τὸ ῥηθέν· φύσιν δὲ ἔχει ὅσα

nature on the one hand therefore is the said; nature but has as many

τοιαύτην ἔχει ἀρχήν. καὶ ἔστιν πάντα ταῦτα οὐσία· ὑποκείμενον

such has a principle. and there is all these essence; substrate

γάρ τι, καὶ ἐν ὑποκειμένῳ ἐστὶν ἡ φύσις ἀεί.

for something, and in the substrate is the nature always.

κατὰ φύσιν δὲ ταῦτά τε καὶ ὅσα τούτοις ὑπάρχει

according to nature but these and also all that pertains to these exists

καθ' αὑτά, οἷον τῷ πυρὶ φέρεσθαι ἄνω· τοῦτο γὰρ

according to themselves, as to the fire to be borne upwards; this for

φύσις μὲν οὐκ [193a] ἔστιν οὐδ' ἔχει φύσιν, φύσει δὲ

nature on the one hand not [193a] is nor has nature, by nature but

καὶ κατὰ φύσιν ἐστίν.

and according to nature it is.

τί μὲν οὖν ἐστιν ἡ φύσις, εἴρηται, καὶ τί τὸ φύσει

what on one hand therefore is the nature, has been said, and what the by nature

καὶ κατὰ φύσιν. ὡς δ' ἔστιν ἡ φύσις, πειρᾶσθαι δεικνύναι

and according to nature. as but it is the nature, to attempt to demonstrate

γελοῖον· φανερὸν γὰρ ὅτι τοιαῦτα τῶν ὄντων ἐστὶν πολλά. τὸ

is ridiculous; evident for that such things of the beings there are many. the

δὲ δεικνύναι τὰ φανερὰ διὰ τῶν ἀφανῶν οὐ δυναμένου κρίνειν

but to demonstrate the evident through the hidden not being able to discern

ἐστὶ τὸ δι' αὑτὸ καὶ μὴ δι' αὑτὸ γνώριμον (ὅτι δ' ἐνδέχεται τοῦτο

is the by itself and not by itself recognizable (that but it may this

πάσχειν, οὐκ ἄδηλον· συλλογίσαιτο γὰρ ἄν τις ἐκ γενετῆς ὢν

suffer, not unclear; let him consider for if anyone from birth being

τυφλὸς περὶ χρωμάτων), ὥστε ἀνάγκη τοῖς τοιούτοις περὶ τῶν

blind about colors), so that necessity for those such regarding the

ὀνομάτων εἶναι τὸν λόγον, νοεῖν δὲ μηδέν.

names to be the account, to think but nothing.

δοκεῖ δ' ἡ φύσις καὶ ἡ οὐσία τῶν φύσει ὄντων ἐνίοις εἶναι τὸ

it seems (and) the nature and the essence of the beings existing in some to be the

πρῶτον ἐνυπάρχον ἑκάστῳ, ἀρρύθμιστον <ὂν>καθ' ἑαυτό, οἷον

first existing in each, uncountable being according to itself, such as

κλίνης φύσις τὸ ξύλον, ἀνδριάντος δ' ὁ χαλκός.

a bed's nature the wood, of a statue and the bronze.

σημεῖον δέ φησιν Ἀντιφῶν ὅτι, εἴ τις κατορύξειε κλίνην καὶ

A sign but says Antiphon that, if someone should dig a bed and

λάβοι δύναμιν ἡ σηπεδὼν ὥστε ἀνεῖναι βλαστόν, οὐκ ἂν γενέσθαι

take power the decay so as to produce a shoot, not would come to be

κλίνην ἀλλὰ ξύλον, ὡς τὸ μὲν κατὰ συμβεβηκὸς ὑπάρχον, τὴν

a bed but wood, as the indeed according to accidental existence, the

κατὰ νόμον διάθεσιν καὶ τὴν τέχνην, τὴν δ' οὐσίαν οὖσαν

according to natural arrangement and the art, the and essence being

ἐκείνην ἣ καὶ διαμένει ταῦτα πάσχουσα συνεχῶς.

that which and remains these undergoing continuously.

εἰ δὲ καὶ τούτων ἕκαστον πρὸς ἕτερόν τι ταὐτὸ τοῦτο πέπονθεν

if but and of these each towards another something the same this has experienced

(οἷον ὁ μὲν χαλκὸς καὶ ὁ χρυσὸς πρὸς ὕδωρ, τὰ δ' ὀστᾶ

(as the on the one hand copper and the gold towards water, the but bones

καὶ ξύλα πρὸς γῆν, ὁμοίως δὲ καὶ τῶν ἄλλων ὁτιοῦν), ἐκεῖνο

and wood towards earth, similarly but also of the others of any kind), that

τὴν φύσιν εἶναι καὶ τὴν οὐσίαν αὐτῶν. διόπερ οἱ μὲν πῦρ,

the nature to be and the essence of them. Therefore the on the one hand fire,

οἱ δὲ γῆν, οἱ δ' ἀέρα φασίν, οἱ δὲ ὕδωρ, οἱ δ' ἔνια

the on the other hand earth, the but air they say, the whereas water, the but some

τούτων, οἱ δὲ πάντα ταῦτα τὴν φύσιν εἶναι τὴν τῶν ὄντων. ὃ

of these, the whereas all these the nature to be the of the beings. that

γάρ τις αὐτῶν ὑπέλαβε τοιοῦτον, εἴτε ἓν εἴτε πλείω, τοῦτο καὶ

for someone of them has assumed such, whether one or more, this and

τοσαῦτά φησιν εἶναι τὴν ἅπασαν οὐσίαν, τὰ δὲ ἄλλα πάντα πάθη

these says to be the total substance, the but other all states

τούτων καὶ ἕξεις καὶ διαθέσεις, καὶ τούτων μὲν ὁτιοῦν

of these and dispositions and arrangements, and of these on one hand whatever

ἀΐδιον (οὐ γὰρ εἶναι μεταβολὴν αὐτοῖς ἐξ αὑτῶν), τὰ δ' ἄλλα

eternal (not for to be a change for them from themselves), the however other

γίγνεσθαι καὶ φθείρεσθαι ἀπειράκις.

to come into being and to perish infinitely.

ἕνα μὲν οὖν τρόπον οὕτως ἡ φύσις λέγεται, ἡ πρώτη
one on one hand therefore way like this the nature is called, the first
ἑκάστῳ ὑποκειμένη ὕλη τῶν ἐχόντων ἐν αὑτοῖς ἀρχὴν
to each underlying matter of those having in themselves a principle
κινήσεως καὶ μεταβολῆς,
of movement and of change,

ἄλλον δὲ τρόπον ἡ μορφὴ καὶ τὸ εἶδος τὸ κατὰ τὸν λόγον.
another but way the form and the essence the according to the reason.

ὥσπερ γὰρ τέχνη λέγεται τὸ κατὰ τέχνην καὶ τὸ τεχνικόν, οὕτω
just as for art is called the according to art and the technical, so
καὶ φύσις τὸ κατὰ φύσιν [λέγεται] καὶ τὸ φυσικόν, οὔτε δὲ
and nature the according to nature [is called] and the natural, neither but
ἐκεῖ πω φαῖμεν ἂν ἔχειν κατὰ τὴν τέχνην οὐδέν, εἰ δυνάμει
there at all we say if to have according to the art nothing, if in potential
μόνον ἐστὶ κλίνη, μή πω δ' ἔχει τὸ εἶδος τῆς κλίνης, οὐδ'
only is a bed, not at all but has the form of the bed, nor
εἶναι τέχνην, οὔτ' ἐν τοῖς φύσει συνισταμένοις· τὸ γὰρ
to be an art, nor in the things of nature that are constituted; for the for
δυνάμει σὰρξ ἢ ὀστοῦν οὔτ' [193b] ἔχει πω τὴν ἑαυτοῦ φύσιν,
in potential flesh or bone nor [193b] has at all the its nature,
πρὶν ἂν λάβῃ τὸ εἶδος τὸ κατὰ τὸν λόγον, ᾧ ὁριζόμενοι
before it receives the form the according to the account, which we define

λέγομεν τί ἐστι σὰρξ ἢ ὀστοῦν, οὔτε φύσει ἐστίν. ὥστε ἄλλον

we say what is flesh or of bone, nor by nature it is. Therefore another

τρόπον ἡ φύσις ἂν εἴη τῶν ἐχόντων ἐν αὐτοῖς κινήσεως

way the nature would be of those possessing in themselves movement

ἀρχὴν ἡ μορφὴ καὶ τὸ εἶδος, οὐ χωριστὸν ὂν ἀλλ' ἢ

a principle the form and the essence, not separate being but rather

κατὰ τὸν λόγον. (τὸ δ' ἐκ τούτων φύσις μὲν οὐκ ἔστιν,

according to the reason. (the and of these nature indeed is not, it exists,

φύσει δέ, οἷον ἄνθρωπος.)

by nature, however, such as a human.)

καὶ μᾶλλον αὕτη φύσις τῆς ὕλης· ἕκαστον γὰρ τότε λέγεται

And more this nature of the matter. Each indeed then is said

ὅταν ἐντελεχείᾳ ᾖ, μᾶλλον ἢ ὅταν δυνάμει. ἔτι γίγνεται

when in actuality it is, rather than when potentially. Moreover, it becomes

ἄνθρωπος ἐξ ἀνθρώπου, ἀλλ' οὐ κλίνη ἐκ κλίνης· διὸ

a human from a human, but not a bed from a bed; therefore

καί φασιν οὐ τὸ σχῆμα εἶναι τὴν φύσιν ἀλλὰ τὸ ξύλον, ὅτι

also they say not the shape to be the nature but the wood, because

γένοιτ' ἄν, εἰ βλαστάνοι, οὐ κλίνη ἀλλὰ ξύλον. εἰ δ' ἄρα

let it be if if it were to sprout, not incline but wood. if then consequently

τοῦτο φύσις, καὶ ἡ μορφὴ φύσις· γίγνεται γὰρ ἐξ ἀνθρώπου

this, nature, and the form of nature; it becomes for out of man

ἄνθρωπος.

man.

ἔτι δ' ἡ φύσις ἡ λεγομένη ὡς γένεσις ὁδός ἐστιν εἰς φύσιν. οὐ

Yet and the nature the called as generation way is into nature. not

γὰρ ὥσπερ ἡ ἰάτρευσις λέγεται οὐκ εἰς ἰατρικὴν ὁδὸς ἀλλ' εἰς

for just as the healing is called not into medical way but into

ὑγίειαν· ἀνάγκη μὲν γὰρ ἀπὸ ἰατρικῆς οὐκ εἰς ἰατρικὴν εἶναι τὴν

health· necessity indeed for from medical not into medical to be the

ἰάτρευσιν, οὐχ οὕτω δ' ἡ φύσις ἔχει πρὸς τὴν φύσιν, ἀλλὰ τὸ

healing, not thus and the nature has towards the nature, but the

φυόμενον ἐκ τινὸς εἰς τὶ ἔρχεται ἢ φύεται. τί οὖν

growing from something into something comes in which it grows. what then

φύεται; οὐχὶ ἐξ οὗ, ἀλλ' εἰς ὅ. ἡ ἄρα μορφὴ φύσις.

grows? not from which, but into that. the therefore form nature.

ἡ δὲ μορφὴ καὶ ἡ φύσις διχῶς λέγεται· καὶ γὰρ ἡ στέρησις

the but form and the nature in two ways is said· and for the deprivation

εἴδός πώς ἐστιν. εἰ δ' ἔστιν στέρησις καὶ ἐναντίον τι περὶ

of form how it is. if but it is deprivation and opposite something concerning

τὴν ἁπλῆν γένεσιν ἢ μὴ ἔστιν, ὕστερον ἐπισκεπτέον.

the simple essence or not it is, later it must be examined.

Κεφάλαιο 2

Chapter 2

Ἐπεὶ δὲ διώρισται ποσαχῶς ἡ φύσις, μετὰ τοῦτο θεωρητέον

Since but it has been defined in various ways, the nature, after this it must be considered

τίνι διαφέρει ὁ μαθηματικὸς τοῦ φυσικοῦ (καὶ γὰρ ἐπίπεδα καὶ

what differs the mathematical of the natural (and for flat and

στερεὰ ἔχει τὰ φυσικὰ σώματα καὶ μήκη καὶ στιγμάς, περὶ ὧν

solid has the natural bodies and lengths and points, regarding which

σκοπεῖ ὁ μαθηματικός)·

the mathematician considers the mathematician).

ἔτι εἰ ἡ ἀστρολογία ἑτέρα ἢ μέρος τῆς φυσικῆς· εἰ γὰρ τοῦ
Also if the astrology another or part of the physics. if for of the
φυσικοῦ τὸ τί ἐστιν ἥλιος ἢ σελήνη εἰδέναι, τῶν δὲ συμβεβηκότων
natural the what is sun or moon to know, of the but occurrences
καθ' αὐτὰ μηδέν, ἄτοπον, ἄλλως τε καὶ ὅτι φαίνονται λέγοντες
according to itself nothing, absurd, otherwise and and that appear saying
οἱ περὶ φύσεως καὶ περὶ σχήματος σελήνης καὶ ἡλίου, καὶ
the concerning nature and concerning shape of the moon and of the sun, and
δὴ καὶ πότερον σφαιροειδὴς ἡ γῆ καὶ ὁ κόσμος ἢ οὔ.
indeed and whether spherical the earth and the cosmos or not.

περὶ τούτων μὲν οὖν πραγματεύεται καὶ ὁ μαθηματικός, ἀλλ'
about these indeed therefore discusses and the mathematician, but
οὐχ ᾗ φυσικοῦ σώματος πέρας ἕκαστον· οὐδὲ τὰ συμβεβηκότα
not in which of the physical body limit each one. nor the accidents
θεωρεῖ ᾗ τοιούτοις οὖσι συμβέβηκεν· διὸ καὶ χωρίζει·
he observes in which such being has occurred. therefore and it separates;
χωριστὰ γὰρ τῇ νοήσει κινήσεώς ἐστι, καὶ οὐδὲν διαφέρει, οὐδὲ
separations for the intellect of motion is, and nothing differs, nor
γίγνεται ψεῦδος χωριζόντων. λανθάνουσι δὲ τοῦτο ποιοῦντες
does it become a falsehood of those separating. they are concealed but this doing
καὶ οἱ τὰς ἰδέας λέγοντες· τὰ γὰρ φυσικὰ [194a] χωρίζουσιν
and the the ideas saying; the for physical [194a] separate
ἧττον ὄντα χωριστὰ τῶν μαθηματικῶν. γίγνοιτο δ' ἂν τοῦτο
less being separations of the mathematical. it would become but if this
δῆλον, εἴ τις ἑκατέρων πειρῷτο λέγειν τοὺς ὅρους, καὶ
is clear, if anyone of each were to attempt to say the definitions, and

αὐτῶν καὶ τῶν συμβεβηκότων. τὸ μὲν γὰρ περιττὸν ἔσται
of themselves and of the accidents. the indeed for the excess will be
καὶ τὸ ἄρτιον καὶ τὸ εὐθὺ καὶ τὸ καμπύλον, ἔτι δὲ ἀριθμὸς
and the appropriate and the straight and the curved, still but number
καὶ γραμμὴ καὶ σχῆμα, ἄνευ κινήσεως, σὰρξ δὲ καὶ ὀστοῦν καὶ
and line and shape, without motion, flesh but and bone and
ἄνθρωπος οὐκέτι, ἀλλὰ ταῦτα ὥσπερ ῥὶς σιμὴ ἀλλ' οὐχ ὡς
human no longer, but these just as a point of likeness but not as
τὸ καμπύλον λέγεται.
the curved is said.

δηλοῖ δὲ καὶ τὰ φυσικώτερα τῶν μαθημάτων, οἷον ὀπτικὴ καὶ
it indicates but and the more natural of the sciences, such as optics and
ἁρμονικὴ καὶ ἀστρολογία· ἀνάπαλιν γὰρ τρόπον τιν' ἔχουσιν τῇ
harmonics and astrology. Again, for a way certain they have in the
γεωμετρίᾳ. ἡ μὲν γὰρ γεωμετρία περὶ γραμμῆς φυσικῆς σκοπεῖ,
geometry. the indeed for geometry concerns the line of nature looks at,
ἀλλ' οὐχ ᾗ φυσική, ἡ δ' ὀπτικὴ μαθηματικὴν μὲν γραμμήν,
but not as natural, the whereas optics mathematical indeed line,
ἀλλ' οὐχ ᾗ μαθηματικὴ ἀλλ' ᾗ φυσική.
but not as mathematical but as natural.

ἐπεὶ δ' ἡ φύσις διχῶς, τό τε εἶδος καὶ ἡ ὕλη, ὡς ἂν εἰ
since but the nature distinctly, the and form and the matter, as one might if
περὶ σιμότητος σκοποῖμεν τί ἐστιν, οὕτω θεωρητέον· ὥστ'
concerning similarity we consider what it is, thus must be examined. Therefore,
οὔτ' ἄνευ ὕλης τὰ τοιαῦτα οὔτε κατὰ τὴν ὕλην. καὶ γὰρ
neither without matter the such things nor according to the matter. And indeed

δὴ καὶ περὶ τούτου ἀπορήσειεν ἄν τις, ἐπεὶ δύο αἱ
certainly and concerning this one would wonder if anyone, since two are
φύσεις, περὶ ποτέρας τοῦ φυσικοῦ. ἢ περὶ τοῦ ἐξ ἀμφοῖν;
natures, concerning which of the natural. Or concerning of the from both?
ἀλλ' εἰ περὶ τοῦ ἐξ ἀμφοῖν, καὶ περὶ ἑκατέρας. πότερον οὖν
But if concerning of the from both, and concerning each. Which one then
τῆς αὐτῆς ἢ ἄλλης ἑκατέραν γνωρίζειν;
of the same or another's each one to know?

εἰς μὲν γὰρ τοὺς ἀρχαίους ἀποβλέψαντι δόξειεν ἂν εἶναι τῆς
For indeed for the ancients looking back it would seem that to be of the
ὕλης (ἐπὶ μικρὸν γάρ τι μέρος Ἐμπεδοκλῆς καὶ Δημόκριτος τοῦ
matter upon small for some part Empedocles and Democritus of the
εἴδους καὶ τοῦ τί ἦν εἶναι ἥψαντο)·
form and of the what was to be they touched)

εἰ δὲ ἡ τέχνη μιμεῖται τὴν φύσιν, τῆς δὲ αὐτῆς ἐπιστήμης εἰδέναι
if but the art imitates the nature, of the but of itself science to know
τὸ εἶδος καὶ τὴν ὕλην μέχρι του (οἷον ἰατροῦ ὑγίειαν καὶ χολὴν
the form and the matter up to that such as of a doctor health and bile
καὶ φλέγμα, ἐν οἷς ἡ ὑγίεια, ὁμοίως δὲ καὶ οἰκοδόμου τό τε εἶδος
and phlegm, in which the health, similarly but and of a builder the and form
τῆς οἰκίας καὶ τὴν ὕλην, ὅτι πλίνθοι καὶ ξύλα· ὡσαύτως δὲ
of the house and the matter, since bricks and wood; likewise but
καὶ ἐπὶ τῶν ἄλλων), καὶ τῆς φυσικῆς ἂν εἴη τὸ γνωρίζειν
and upon the others), and of the physical if it would be the knowledge
ἀμφοτέρας τὰς φύσεις.
of both the natures.

ἔτι τὸ οὗ ἕνεκα καὶ τὸ τέλος τῆς αὐτῆς, καὶ ὅσα
still the of which for the sake of and the end of the itself, and as many as
τούτων ἕνεκα. ἡ δὲ φύσις τέλος καὶ οὗ ἕνεκα
of these for the sake of. the but nature end and of which for the sake of
(ὧν γὰρ συνεχοῦς τῆς κινήσεως οὔσης ἔστι τι τέλος,
(of which for continuous of the movement being it is something end,
τοῦτο <τὸ> ἔσχατον καὶ τὸ οὗ ἕνεκα· διὸ καὶ ὁ
this <the> ultimate and the of which for the sake of; therefore and the
ποιητὴς γελοίως προήχθη εἰπεῖν "ἔχει τελευτήν, ἥσπερ οὕνεκ'
poet humorously led forth to say "has an end, just as for the sake of which
ἐγένετο"· βούλεται γὰρ οὐ πᾶν εἶναι τὸ ἔσχατον τέλος, ἀλλὰ τὸ
it came to be"· he wishes for not every to be the ultimate end, but the
βέλτιστον)· ἐπεὶ καὶ ποιοῦσιν αἱ τέχναι τὴν ὕλην αἱ μὲν
best)· since also they create the arts the matter the on one hand
ἁπλῶς αἱ δὲ εὐεργόν, καὶ χρώμεθα ὡς ἡμῶν ἕνεκα πάντων
simply the but beneficial, and we make use of as of us for the sake of all
ὑπαρχόντων (ἐσμὲν γάρ πως καὶ ἡμεῖς τέλος· διχῶς γὰρ τὸ
existing (we are for how and we finally; separately for the
οὗ ἕνεκα· εἴρηται δ' ἐν τοῖς περὶ φιλοσοφίας).
of which because it has been stated but in the concerning philosophy).
δύο [194b] δὲ αἱ ἄρχουσαι τῆς ὕλης καὶ γνωρίζουσαι τέχναι,
two [194b] and the ruling of the matter and knowing arts,
ἥ τε χρωμένη καὶ τῆς ποιητικῆς ἡ ἀρχιτεκτονική. διὸ καὶ
which and using and of the poetic the architecture. therefore and
ἡ χρωμένη ἀρχιτεκτονική πως, διαφέρει δὲ ᾗ ἡ μὲν
the using architecture somehow, differs but in that the on one hand

τοῦ εἴδους γνωριστική, ἡ ἀρχιτεκτονική, ἡ δὲ ὡς

of the form is informative, the architecture, the on the other hand as

ποιητική, τῆς ὕλης· ὁ μὲν γὰρ κυβερνήτης ποῖόν τι τὸ

poetic, of the matter; the on one hand for the helmsman what kind of the

εἶδος τοῦ πηδαλίου γνωρίζει καὶ ἐπιτάττει, ὁ δ' ἐκ ποίου ξύλου

form of the rudder knows and commands, the but from what wood

καὶ ποίων κινήσεων ἔσται. ἐν μὲν οὖν τοῖς κατὰ

and what kind of movements it will be. in indeed therefore the according to

τέχνην ἡμεῖς ποιοῦμεν τὴν ὕλην τοῦ ἔργου ἕνεκα, ἐν δὲ

art we create the material of the work for the sake of, in but

τοῖς φυσικοῖς ὑπάρχει οὖσα.

the natural exists being.

ἔτι τῶν πρός τι ἡ ὕλη· ἄλλῳ γὰρ εἴδει ἄλλη ὕλη. μέχρι

yet of the to something the matter; to another for kind another matter. until

δὴ πόσου τὸν φυσικὸν δεῖ εἰδέναι τὸ εἶδος καὶ τὸ τί ἐστιν; ἢ

indeed how much the physical must know the form and the what it is? or

ὥσπερ ἰατρὸν νεῦρον ἢ χαλκέα χαλκόν, μέχρι τοῦ τίνος [γὰρ]

just as a physician a nerve or bronze copper, until of what kind [for]

ἕνεκα ἕκαστον, καὶ περὶ ταῦτα ἅ ἐστι χωριστὰ μὲν εἴδει,

sake of each, and about these which is separate on the one hand, kinds,

ἐν ὕλῃ δέ; ἄνθρωπος γὰρ ἄνθρωπον γεννᾷ καὶ ἥλιος.

in matter on the other hand? man for man begets and the sun.

πῶς δ' ἔχει τὸ χωριστὸν καὶ τί ἐστι, φιλοσοφίας ἔργον διορίσαι

how but has the separate and what is, of philosophy work to delineate

τῆς πρώτης.

of the first.

Κεφάλαιο 3

Chapter 3

Διωρισμένων δὲ τούτων ἐπισκεπτέον περὶ τῶν αἰτίων, ποῖά
Of the defined but these to be examined about the causes, what kind
τε καὶ πόσα τὸν ἀριθμόν ἐστιν. ἐπεὶ γὰρ τοῦ εἰδέναι χάριν
as well as and how many the number is. since for of the to know thank
ἡ πραγματεία, εἰδέναι δὲ οὐ πρότερον οἰόμεθα ἕκαστον πρὶν ἂν
the discourse, to know but not beforehand we think each before we may
λάβωμεν τὸ διὰ τί περὶ ἕκαστον (τοῦτο δ' ἐστὶ τὸ λαβεῖν
take the through what concerning each (this indeed is the taking
τὴν πρώτην αἰτίαν), δῆλον ὅτι καὶ ἡμῖν τοῦτο ποιητέον καὶ περὶ
the first cause), clear that also to us this must be done and about
γενέσεως καὶ φθορᾶς καὶ πάσης τῆς φυσικῆς μεταβολῆς, ὅπως
generation and corruption and all of the physical change, so that
εἰδότες αὐτῶν τὰς ἀρχὰς ἀνάγειν εἰς αὐτὰς πειρώμεθα τῶν
knowing them the principles we may derive to them let us attempt of the
ζητουμένων ἕκαστον. ἕνα μὲν οὖν τρόπον αἴτιον λέγεται τὸ
sought each. one indeed therefore manner cause is said the
ἐξ οὗ γίγνεταί τι ἐνυπάρχοντος, οἷον ὁ χαλκὸς τοῦ
from which arises something existing, such as the bronze of the
ἀνδριάντος καὶ ὁ ἄργυρος τῆς φιάλης καὶ τὰ τούτων γένη·
statue and the silver of the of the vase and the of these kinds;

ἄλλον δὲ τὸ εἶδος καὶ τὸ παράδειγμα, τοῦτο δ' ἐστὶν ὁ λόγος ὁ
another but the form and the example, this but is the reason the
τοῦ τί ἦν εἶναι καὶ τὰ τούτου γένη (οἷον τοῦ διὰ πασῶν τὰ
of the what was to be and the of this kinds (such as of through all the

δύο πρὸς ἕν, καὶ ὅλως ὁ ἀριθμός) καὶ τὰ μέρη τὰ ἐν τῷ λόγῳ.
two towards one, and generally the number) and the parts the in the reason.

ἔτι ὅθεν ἡ ἀρχὴ τῆς μεταβολῆς ἡ πρώτη ἢ τῆς ἠρεμήσεως,
further whence the beginning of change the first or of rest,
οἷον ὁ βουλεύσας αἴτιος, καὶ ὁ πα τὴρ τοῦ τέκνου, καὶ ὅλως
such as the deliberator cause, and the father of the child, and generally
τὸ ποιοῦν τοῦ ποιουμένου καὶ τὸ μεταβάλλον τοῦ μεταβαλλομένου.
the making of the being made and the changing of the changing.

ἔτι ὡς τὸ τέλος· τοῦτο δ' ἐστὶν τὸ οὗ ἕνεκα, οἷον τοῦ
further as the end; this but is the which for the sake of, such as the
περιπατεῖν ἡ ὑγίεια· διὰ τί γὰρ περιπατεῖ; φαμέν "ἵνα
walking the health; through why for does (one) walk? we say "in order that
ὑγιαίνῃ", καὶ εἰπόντες οὕτως οἰόμεθα ἀποδεδωκέναι τὸ
(it) may be healthy", and having said thus we think to have given the
αἴτιον. καὶ ὅσα δὴ κινήσαντος ἄλλου μεταξὺ γίγνεται τοῦ
cause. And as many as indeed of (the one) moving another in between happens the
τέλους, οἷον τῆς ὑγιείας ἡ ἰσχνασία ἢ ἡ [195a] κάθαρσις ἢ τὰ
end, such as the health the thinness or the [195a] purification or the
φάρμακα ἢ τὰ ὄργανα· πάντα γὰρ ταῦτα τοῦ τέλους ἕνεκά
medicines or the instruments; all for these the end for the sake of
ἐστιν, διαφέρει δὲ ἀλλήλων ὡς ὄντα τὰ μὲν ἔργα τὰ δ'
exist, differs but from one another as beings the indeed works the but
ὄργανα.
instruments.

τὰ μὲν οὖν αἴτια σχεδὸν τοσαυταχῶς λέγεται,
the indeed therefore causes almost in such a great number is said,

συμβαίνει δὲ πολλαχῶς λεγομένων τῶν αἰτίων καὶ πολλὰ τοῦ αὐτοῦ

happens and in many ways of things said of the causes and many of the same

αἴτια εἶναι, οὐ κατὰ συμβεβηκός, οἷον τοῦ ἀνδριάντος

cause to be, not according to that which has happened, such as of the statue

καὶ ἡ ἀνδριαντοποιικὴ καὶ ὁ χαλκός, οὐ καθ' ἕτερόν

and the sculpture and the bronze, not in accordance with different

τι ἀλλ' ᾗ ἀνδριάς, ἀλλ' οὐ τὸν αὐτὸν τρόπον, ἀλλὰ τὸ

something but which statue, but not the same manner, but the

μὲν ὡς ὕλη τὸ δ' ὡς ὅθεν ἡ κίνησις. ἔστιν δέ τινα

on one hand as matter the and as from where the motion. There is but some

καὶ ἀλλήλων αἴτια, οἷον τὸ πονεῖν τῆς εὐεξίας καὶ αὕτη τοῦ

and of each other causes, such as the pain of the well-being and this of the

πονεῖν· ἀλλ' οὐ τὸν αὐτὸν τρόπον, ἀλλὰ τὸ μὲν ὡς τέλος τὸ

pain· but not the same manner, but the on one hand as end the

δ' ὡς ἀρχὴ κινήσεως. ἔτι δὲ τὸ αὐτὸ τῶν ἐναντίων ἐστίν·

and as beginning of motion. Furthermore and the same of the opposers it is;

ὃ γὰρ παρὸν αἴτιον τοῦδε, τοῦτο καὶ ἀπὸν αἰτιώμεθα ἐνίοτε τοῦ

what for present cause of this, this and absent we are blamed sometimes of the

ἐναντίου, οἷον τὴν ἀπουσίαν τοῦ κυβερνήτου τῆς τοῦ πλοίου

opposer, such as the absence of the helmsman of the the ship

ἀνατροπῆς, οὗ ἦν ἡ παρουσία αἰτία τῆς σωτηρίας.

overturning, of which there was the presence cause of the salvation.

ἅπαντα δὲ τὰ νῦν εἰρημένα αἴτια εἰς τέτταρας πίπτει τρόπους τοὺς

all but the now stated causes into four falls ways the

φανερωτάτους. τὰ μὲν γὰρ στοιχεῖα τῶν συλλαβῶν καὶ ἡ

most evident. the on one hand for elements of the syllables and the

ὕλη τῶν σκευαστῶν καὶ τὸ πῦρ καὶ τὰ τοιαῦτα τῶν σωμάτων

matter of the compounds and the fire and the such things of the bodies

καὶ τὰ μέρη τοῦ ὅλου καὶ αἱ ὑποθέσεις τοῦ συμπεράσματος ὡς

and the parts of the whole and the hypotheses of the conclusion as

τὸ ἐξ οὗ αἰτιά ἐστιν, τούτων δὲ τὰ μὲν ὡς τὸ ὑποκείμενον,

the from which cause is, of these but the on one hand as the subject,

οἷον τὰ μέρη, τὰ δὲ ὡς τὸ τί ἦν εἶναι, τό τε ὅλον

such as the parts, the on the other hand as the what it was to be, the and whole

καὶ ἡ σύνθεσις καὶ τὸ εἶδος· τὸ δὲ σπέρμα καὶ ὁ ἰατρὸς καὶ

and the composition and the form. the but seed and the doctor and

ὁ βουλεύσας καὶ ὅλως τὸ ποιοῦν, πάντα ὅθεν ἡ ἀρχὴ

the having deliberated and altogether the making, all whence the beginning

τῆς μεταβολῆς ἢ στάσεως [ἢ κινήσεως]· τὰ δ' ὡς τὸ τέλος καὶ

of the change or of state or of motion. the but as the end and

τἀγαθὸν τῶν ἄλλων· τὸ γὰρ οὗ ἕνεκα βέλτιστον καὶ τέλος

the good of the others. the for of which on account of the best and end

τῶν ἄλλων ἐθέλει εἶναι· διαφερέτω δὲ μηδὲν εἰπεῖν αὐτὸ ἀγαθὸν ἢ

of the others intends to be. let it differ but nothing to say itself good or

φαινόμενον ἀγαθόν.

appearing good.

τὰ μὲν οὖν αἴτια ταῦτα καὶ τοσαῦτά ἐστι τῷ εἴδει·

the on the one hand therefore causes these and so many are to the form.

τρόποι δὲ τῶν αἰτίων ἀριθμῷ μὲν εἰσὶ πολλοί, κεφαλαιούμενοι

ways but of the causes in number on the one hand there are many, summarized

δὲ καὶ οὗτοι ἐλάττους. λέγεται γὰρ αἴτια πολλαχῶς, καὶ αὐτῶν
but and these fewer. it is said for causes in various ways, and of them
τῶν ὁμοειδῶν προτέρως καὶ ὑστέρως ἄλλο ἄλλου, οἷον
of the similar previously and later one another, another, such as
ὑγιείας ἰατρὸς καὶ τεχνίτης, καὶ τοῦ διὰ πασῶν τὸ διπλάσιον
of health physician and craftsman, and of the through all the double
καὶ ἀριθμός, καὶ ἀεὶ τὰ περιέχοντα πρὸς τὰ καθ' ἕκαστον.
and number, and always the containing in relation to the as for each one.
ἔτι δ' ὡς τὸ συμβεβηκὸς καὶ τὰ τούτων γένη, οἷον ἀνδριάντος
also but as the incidental and the of these kinds, such as of a statue
ἄλλως Πολύκλειτος καὶ ἄλλως ἀνδριαντοποιός, ὅτι συμβέβηκε
otherwise Polykleitos and otherwise sculptor, that it has happened
τῷ ἀνδριαντοποιῷ τὸ Πολυκλείτῳ εἶναι. καὶ τὰ περιέχοντα δὲ τὸ
to the sculptor the Polykleitos to be. and the containing but the
συμβεβηκός, οἷον εἰ ὁ ἄνθρωπος αἴτιος εἴη ἀνδριάντος
incidental, such as if the man is responsible would be of a statue
ἢ [195b] ὅλως ζῷον. ἔστι δὲ καὶ τῶν συμβεβηκότων
or [195b] overall a living being. there is but also of the incidents
ἄλλα ἄλλων πορρώτερον καὶ ἐγγύτερον, οἷον εἰ ὁ λευκὸς καὶ
other of others further and closer, such as if the white and
ὁ μουσικὸς αἴτιος λέγοιτο τοῦ ἀνδριάντος.
the musical cause may be said of the statue.

πάντα δὲ καὶ τὰ οἰκείως λεγόμενα καὶ τὰ κατὰ συμβεβηκὸς
everything but and the appropriately said and the according to incidental
τὰ μὲν ὡς δυνάμενα λέγεται τὰ δ' ὡς ἐνεργοῦντα, οἷον τοῦ
the on one hand as capable are said the but as active, such as of the
οἰκοδομεῖσθαι οἰκίαν οἰκοδόμος ἢ οἰκοδομῶν οἰκοδόμος.
being built house builder or of builders builder.

ὁμοίως δὲ λεχθήσεται καὶ ἐφ' ὧν αἴτια τὰ αἴτια τοῖς εἰρημένοις,

similarly but will be said and concerning which causes the causes to the aforementioned,

οἷον τουδὶ τοῦ ἀνδριάντος ἢ ἀνδριάντος ἢ ὅλως εἰκόνος,

such as this of the statue or statue or altogether of image,

καὶ χαλκοῦ τοῦδε ἢ χαλκοῦ ἢ ὅλως ὕλης· καὶ ἐπὶ τῶν

and of bronze of this or of bronze or altogether of matter. And concerning the

συμβεβηκότων ὡσαύτως. ἔτι δὲ συμπλεκόμενα καὶ ταῦτα κἀκεῖνα

occurrences likewise. Moreover but interwoven and these and those

λεχθήσεται, οἷον οὐ Πολύκλειτος οὐδὲ ἀνδριαντοποιός, ἀλλὰ

will be said, such as not Polycleitus nor a sculptor, but

Πολύκλειτος ἀνδριαντοποιός. ἀλλ' ὅμως ἅπαντα ταῦτά ἐστι τὸ

Polycleitus is a sculptor. But however all these is the

μὲν πλῆθος ἕξ, λεγόμενα δὲ διχῶς· ἢ γὰρ ὡς τὸ

on the one hand multitude six, called but in two ways: either for as the

καθ' ἕκαστον, ἢ ὡς τὸ γένος, ἢ ὡς τὸ συμβεβηκός, ἢ ὡς

for each individual, or as the kind, or as the occurrence, or as

τὸ γένος τοῦ συμβεβηκότος, ἢ ὡς συμπλεκόμενα ταῦτα ἢ ὡς

the kind of the occurrence, or as intertwined these or as

ἁπλῶς λεγόμενα· πάντα δὲ ἢ ἐνεργοῦντα ἢ κατὰ δύναμιν.

simply said. all but or acting or according to power.

διαφέρει δὲ τοσοῦτον, ὅτι τὰ μὲν ἐνεργοῦντα καὶ τὰ καθ' ἕκαστον

differs but so much, that the indeed acting and the each individual

ἅμα ἔστι καὶ οὐκ ἔστι καὶ ὧν αἴτια, οἷον ὅδ' ὁ

at the same time is and not is and of which the cause, such as this the

ἰατρεύων τῷδε τῷ ὑγιαζομένῳ καὶ ὅδε ὁ οἰκοδομῶν τῷδε τῷ

healer to this the being healed and this the builder to this the

οἰκοδομουμένῳ, τὰ δὲ κατὰ δύναμιν οὐκ ἀεί. φθείρεται γὰρ
being built, the but according to power not always. is destroyed for
οὐχ ἅμα ἡ οἰκία καὶ ὁ οἰκοδόμος.
not at the same time the house and the builder.

δεῖ δ' ἀεὶ τὸ αἴτιον ἑκάστου τὸ ἀκρότατον ζητεῖν, ὥσπερ
it is necessary but always the cause of each the ultimate to seek, just as
καὶ ἐπὶ τῶν ἄλλων (οἷον ἄνθρωπος οἰκοδομεῖ ὅτι οἰκοδόμος, ὁ
and in the others (such as the person who builds since a builder, the
δ' οἰκοδόμος κατὰ τὴν οἰκοδομικήν· τοῦτο τοίνυν πρότερον
but builder according to the art of building. This therefore previously
τὸ αἴτιον, καὶ οὕτως ἐπὶ πάντων)· ἔτι τὰ μὲν γένη τῶν
the cause, and thus in all.) moreover the on the one hand kinds of
γενῶν, τὰ δὲ καθ' ἕκαστον τῶν καθ' ἕκαστον
generations, the on the other hand according to each of according to each
(οἷον ἀνδριαντοποιὸς μὲν ἀνδριάντος, ὁδὶ δὲ τουδί)·
(such as a sculptor indeed of statues, while on the other hand he here)·
καὶ τὰς μὲν δυνάμεις τῶν δυνατῶν, τὰ δ' ἐνεργοῦντα πρὸς
and the on the one hand powers of the able, the but acting towards
τὰ ἐνεργούμενα.
the acted upon.

ὅσα μὲν οὖν τὰ αἴτια καὶ ὃν τρόπον αἴτια,
As many as on the one hand therefore the causes and in which manner they are causes,
ἔστω ἡμῖν διωρισμένα ἱκανῶς.
let it be for us defined sufficiently.

Κεφάλαιο 4

Chapter 4

Λέγεται δὲ καὶ ἡ τύχη καὶ τὸ αὐτόματον τῶν αἰτίων, καὶ πολλὰ
It is said but and the fortune and the automaton of the causes, and many
καὶ εἶναι καὶ γίγνεσθαι διὰ τύχην καὶ διὰ τὸ αὐτόματον·
and to be and to come to be through fortune and through the automaton.
τίνα οὖν τρόπον ἐν τούτοις ἐστὶ τοῖς αἰτίοις ἡ τύχη καὶ
Which therefore way in these is the causes the fortune and
τὸ αὐτόματον, καὶ πότερον τὸ αὐτὸ ἡ τύχη καὶ τὸ αὐτόματον
the automaton, and whether the same the fortune and the automaton
ἢ ἕτερον, καὶ ὅλως τί ἐστιν ἡ τύχη καὶ τὸ αὐτόματον,
or other, and altogether what is the fortune and the automaton,
ἐπισκεπτέον.
must be examined.

ἔνιοι γὰρ καὶ εἰ ἔστιν ἢ μὴ ἀποροῦσιν· [196a] οὐδὲν γὰρ δὴ
some for and if it is or not they are in difficulty. [196a] nothing for indeed
γίγνεσθαι ἀπὸ τύχης φασίν, ἀλλὰ πάντων εἶναί τι αἴτιον
to happen from chance they say, but of all to be something the cause
ὡρισμένον ὅσα λέγομεν ἀπὸ ταὐτομάτου γίγνεσθαι ἢ τύχης,
defined as many as we say from spontaneity to happen or of chance,
οἷον τοῦ ἐλθεῖν ἀπὸ τύχης εἰς τὴν ἀγοράν, καὶ καταλαβεῖν ὃν
such as the coming from chance into the marketplace, and to catch whom
ἐβούλετο μὲν οὐκ ᾤετο δέ, αἴτιον τὸ βούλεσθαι ἀγοράσαι
he wanted indeed not he thought but, the cause the to want to purchase
ἐλθόντα· ὁμοίως δὲ καὶ ἐπὶ τῶν ἄλλων τῶν ἀπὸ τύχης λεγομένων
having come. similarly and also upon the others of from chance things said

ἀεί τι εἶναι λαβεῖν τὸ αἴτιον, ἀλλ' οὐ τύχην, ἐπεὶ εἴ γέ

always something to be to take the cause, but not chance, since if indeed

τι ἦν ἡ τύχη, ἄτοπον ἂν φανείη ὡς ἀληθῶς, καὶ ἀπορήσειεν

something was the chance, strange might appear as truly, and would wonder

ἄν τις διὰ τί ποτ' οὐδεὶς τῶν ἀρχαίων σοφῶν τὰ αἴτια

if someone through what indeed no one of the ancients wise the causes

περὶ γενέσεως καὶ φθορᾶς λέγων περὶ τύχης οὐδὲν διώρισεν,

regarding of generation and of decay saying about fortune nothing determined,

ἀλλ' ὡς ἔοικεν, οὐδὲν ᾠοντο οὐδ' ἐκεῖνοι εἶναι ἀπὸ τύχης. ἀλλὰ

but as it seems, nothing thought nor those to be from fortune. but

καὶ τοῦτο θαυμαστόν· πολλὰ γὰρ καὶ γίγνεται καὶ ἔστιν ἀπὸ τύχης

also this is wonderful; many for also happens and is from fortune

καὶ ἀπὸ ταὐτομάτου, ἃ οὐκ ἀγνοοῦντες ὅτι ἔστιν ἐπανενεγκεῖν

and from spontaneity, which not being unaware that it is to bring back

ἕκαστον ἐπί τι αἴτιον τῶν γιγνομένων, καθάπερ ὁ παλαιὸς

each upon something cause of the things happening, just as the ancient

λόγος εἶπεν ὁ ἀναιρῶν τὴν τύχην, ὅμως τούτων τὰ μὲν εἶναί

saying said the one who negates the fortune, however of these the indeed are

φασι πάντες ἀπὸ τύχης τὰ δ' οὐκ ἀπὸ τύχης· διὸ καὶ ἁμῶς

they say all from fortune the but not from fortune. therefore and we

γέ πως ἦν ποιητέον αὐτοῖς μνείαν. ἀλλὰ μὴν οὐδ' ἐκείνων

indeed somehow it was to be created for them a mention. but indeed not even of those

γέ τι ᾠοντο εἶναι τὴν τύχην, οἷον φιλίαν ἢ νεῖκος ἢ

indeed anything they thought to be the fortune, like friendship or strife or

νοῦν ἢ πῦρ ἢ ἄλλο γέ τι τῶν τοιούτων. ἄτοπον οὖν

mind or fire or another indeed something of the such things. strange therefore

εἴτε μὴ ὑπελάμβανον εἶναι εἴτε οἰόμενοι παρέλειπον, καὶ

whether not they presumed to be or thinking they were neglecting, and

ταῦτ' ἐνίοτε χρώμενοι, ὥσπερ Ἐμπεδοκλῆς οὐκ ἀεὶ τὸν ἀέρα

these things sometimes using, just as Empedocles not always the air

ἀνωτάτω ἀποκρίνεσθαί φησιν, ἀλλ' ὅπως ἂν τύχῃ. λέγει γοῦν

highest to differentiate, he says, but however it may happen. he says at least

ἐν τῇ κοσμοποιίᾳ ὡς "οὕτω συνέκυρσε θέων τοτέ, πολλάκι

in the cosmology as "thus it has coincided of the gods once, many times

δ' ἄλλως"· καὶ τὰ μόρια τῶν ζῴων ἀπὸ τύχης γενέσθαι τὰ

and otherwise." And the parts of the animals from of chance to become the

πλεῖστά φησιν. εἰσὶ δέ τινες οἳ καὶ τοὐρανοῦ τοῦδε καὶ τῶν

most says. there are but some who and of the heavens of this and of the

κόσμων πάντων αἰτιῶνται τὸ αὐτόματον· ἀπὸ ταὐτομάτου

worlds all are responsible for the spontaneous; from the spontaneous

γὰρ γενέσθαι τὴν δίνην καὶ τὴν κίνησιν τὴν διακρίνασαν καὶ

for to become the vortex and the motion the they distinguished and

καταστήσασαν εἰς ταύτην τὴν τάξιν τὸ πᾶν. καὶ μάλα τοῦτό

established in this the order for the whole. And very this

γε αὐτὸ θαυμάσαι ἄξιον· λέγοντες γὰρ τὰ μὲν ζῷα

indeed it is worthy of admiration is worthy; saying for the indeed animals

καὶ τὰ φυτὰ ἀπὸ τύχης μήτε εἶναι μήτε γίγνεσθαι, ἀλλ' ἤτοι

and the plants from chance neither to be nor to become, but either

φύσιν ἢ νοῦν ἤ τι τοιοῦτον ἕτερον εἶναι τὸ αἴτιον (οὐ γὰρ

nature or mind or something of that sort other to be the cause (not for

ὅ τι ἔτυχεν ἐκ τοῦ σπέρματος ἑκάστου γίγνεται, ἀλλ'

what anything happened from the seed of each it happens, but

ἐκ μὲν τοῦ τοιουδὶ ἐλαία ἐκ δὲ τοῦ τοιουδὶ

from on one hand of the such olive from on the other hand of the such

ἄνθρωπος), τὸν δ' οὐρανὸν καὶ τὰ θειότατα τῶν φανερῶν ἀπὸ

person), the and heaven and the most divine of the visible from

τοῦ αὐτομάτου γενέσθαι, τοιαύτην δ' αἰτίαν μηδεμίαν εἶναι οἵαν

the automatic to become, such and cause none to be such as

τῶν ζώων καὶ τῶν φυτῶν. καίτοι εἰ οὕτως ἔχει, τοῦτ' αὐτὸ ἄξιον

of the animals and of the plants. And yet if so it is, this itself worthy

ἐπιστάσεως, καὶ καλῶς ἔχει λεχ[196b] θῆναί τι περὶ

of investigation, and rightly it is said that there should be something about

αὐτοῦ. πρὸς γὰρ τῷ καὶ ἄλλως ἄτοπον εἶναι τὸ λεγόμενον, ἔτι

it. For indeed to the also otherwise strange to be the said, still

ἀτοπώτερον τὸ λέγειν ταῦτα ὁρῶντας ἐν μὲν τῷ οὐρανῷ

stranger the saying these things seeing in on the one hand the heaven

οὐδὲν ἀπὸ ταὐτομάτου γιγνόμενον, ἐν δὲ τοῖς οὐκ ἀπὸ τύχης πολλὰ

nothing from automatic occurring, in but those not from chance many

συμβαίνοντα ἀπὸ τύχης· καίτοι εἰκός γε ἦν τοὐναντίον

events from chance. And yet it is likely indeed it was the opposite

γίγνεσθαι.

to happen.

εἰσὶ δέ τινες οἷς δοκεῖ εἶναι μὲν αἰτία ἡ τύχη, ἄδηλος δὲ

there are but some to whom it seems to be indeed a cause the fortune, unclear but

ἀνθρωπίνῃ διανοίᾳ ὡς θεῖόν τι οὖσα καὶ δαιμονιώτερον.

human reason as divine something being and more demonic.

ὥστε σκεπτέον καὶ τί ἑκάτερον, καὶ εἰ ταὐτὸν ἢ ἕτερον

So it must be considered and what each one, and if the same or different

τό τε αὐτόματον καὶ ἡ τύχη, καὶ πῶς εἰς τὰ διωρισμένα αἴτια

that both automatic and the fortune, and how into the defined causes

ἐμπίπτουσιν.

fall.

Κεφάλαιο 5

Chapter 5

Πρῶτον μὲν οὖν, ἐπειδὴ ὁρῶμεν τὰ μὲν ἀεὶ ὡσαύτως
First indeed, therefore, since we see the indeed always in the same manner
γιγνόμενα τὰ δὲ ὡς ἐπὶ τὸ πολύ, φανερὸν ὅτι οὐδετέρου τούτων
occurring the but as for the many, it is evident that of neither of these
αἰτία ἡ τύχη λέγεται οὐδὲ τὸ ἀπὸ τύχης, οὔτε τοῦ ἐξ
cause the fortune is spoken of neither the from fortune, nor of the arising
ἀνάγκης καὶ αἰεὶ οὔτε τοῦ ὡς ἐπὶ τὸ πολύ. ἀλλ' ἐπειδὴ ἔστιν
from necessity and always nor of the as at the many. but since it is
ἃ γίγνεται καὶ παρὰ ταῦτα, καὶ ταῦτα πάντες φασὶν εἶναι ἀπὸ
what happens and alongside these, and these all say to be from
τύχης, φανερὸν ὅτι ἔστι τι ἡ τύχη καὶ τὸ αὐτόματον· τά τε
chance, evident that it is something the chance and the automatic; the and
γὰρ τοιαῦτα ἀπὸ τύχης καὶ τὰ ἀπὸ τύχης τοιαῦτα ὄντα ἴσμεν.
for such things from chance and the from chance such things being we know.

τῶν δὲ γιγνομένων τὰ μὲν ἕνεκά του γίγνεται τὰ δ'
of the but things being generated the on one hand for the sake of it is becoming the and
οὔ (τούτων δὲ τὰ μὲν κατὰ προαίρεσιν, τὰ δ' οὐ κατὰ
not (of these but the on one hand according to preference, the and not according to
προαίρεσιν, ἄμφω δ' ἐν τοῖς ἕνεκά του), ὥστε δῆλον ὅτι
preference, both and in the for the sake of it), so that it is clear that
καὶ ἐν τοῖς παρὰ τὸ ἀναγκαῖον καὶ τὸ ὡς ἐπὶ τὸ πολὺ
also in the besides the necessary and the as for the most part the much
ἔστιν ἕνια περὶ ἃ ἐνδέχεται ὑπάρχειν τὸ ἕνεκά του. ἔστι
is some concerning what may exist to exist the for the sake of it. it is

δ’ ἕνεκά του ὅσα τε ἀπὸ διανοίας ἂν πραχθείη καὶ

and for the sake of it as many as and from reasoning if it would be done and

ὅσα ἀπὸ φύσεως. τὰ δὴ τοιαῦτα ὅταν κατὰ συμβεβηκὸς

as many as from nature. the indeed such whenever according to coincidentally

γένηται, ἀπὸ τύχης φαμὲν εἶναι (ὥσπερ γὰρ καὶ ὄν ἐστι τὸ

it happens, from chance we say to be (as for also being is the

μὲν καθ’ αὑτὸ τὸ δὲ κατὰ συμβεβηκός,

on the one hand according to itself the on the other hand in accordance with the accidental,

οὕτω καὶ αἴτιον ἐνδέχεται εἶναι, οἷον οἰκίας καθ’ αὑτὸ

thus and cause may be, to be, such as houses according to itself

μὲν αἴτιον τὸ οἰκοδομικόν, κατὰ συμβεβηκὸς δὲ

on the one hand cause the architectural, in accordance with the accidental but

τὸ λευκὸν ἢ τὸ μουσικόν· τὸ μὲν οὖν καθ’ αὑτὸ

the white or the musical. the on the one hand therefore according to itself

αἴτιον ὡρισμένον, τὸ δὲ κατὰ συμβεβηκὸς ἀόριστον·

cause defined, the but in accordance with the accidental indefinite;

ἄπειρα γὰρ ἂν τῷ ἑνὶ συμβαίη). καθάπερ οὖν ἐλέχθη,

countless for one might in the one happen). just as therefore it was said,

ὅταν ἐν τοῖς ἕνεκά του γιγνομένοις τοῦτο γένηται, τότε λέγεται

when in the names of them arising, this occurs, then is said

ἀπὸ ταὐ τομάτου καὶ ἀπὸ τύχης (αὐτῶν δὲ πρὸς ἄλληλα

from the same segment and from chance (of them but towards one another

τὴν διαφορὰν τούτων ὕστερον διοριστέον· νῦν δὲ τοῦτο ἔστω

the difference of these later to be defined. Now but this let it be

φανερόν, ὅτι ἄμφω ἐν τοῖς ἕνεκά τού ἐστιν)·

evident, that both in the sake of this it is).

οἷον ἕνεκα τοῦ ἀπολαβεῖν τὸ ἀργύριον ἦλθεν ἂν κομιζομένου
such as for the sake of the receiving the silver he came perhaps taking
τὸν ἔρανον, εἰ ᾔδει· ἦλθε δ' οὐ τούτου ἕνεκα, ἀλλὰ συνέβη
the donation, if he knew; he came but not for this sake, but it happened
αὐτῷ ἐλθεῖν, καὶ ποιῆσαι τοῦτο τοῦ κομίσασθαι ἕνεκα· τοῦτο
to him to come, and to do this for the taking sake; this
δὲ οὔθ' ὡς ἐπὶ τὸ πολὺ φοιτῶν εἰς τὸ [197a] χωρίον οὔτ'
but neither as regarding the many visiting to the [197a] place nor
ἐξ ἀνάγκης· ἔστι δὲ τὸ τέλος, ἡ κομιδή, οὐ τῶν ἐν αὐτῷ
from necessity; it is but the end, the act of taking, not of those in it
αἰτίων, ἀλλὰ τῶν προαιρετῶν καὶ ἀπὸ διανοίας· καὶ λέγεταί
responsible, but of those preferred and from thought; and it is said
γε τότε ἀπὸ τύχης ἐλθεῖν, εἰ δὲ προελόμενος καὶ τούτου ἕνεκα ἢ
indeed then from chance to come, if but choosing and for this sake or
ἀεὶ φοιτῶν ἢ ὡς ἐπὶ τὸ πολύ [κομιζόμενος], οὐκ ἀπὸ τύχης.
always visiting or as regarding the much being conveyed not from chance.

δῆλον ἄρα ὅτι ἡ τύχη αἰτία κατὰ συμβεβηκὸς ἐν τοῖς
it is evident therefore that the chance cause according to coincidence in the
κατὰ προαίρεσιν τῶν ἕνεκά του. διὸ περὶ τὸ αὐτὸ
according to preferences of the sake of it. therefore regarding the same
διάνοια καὶ τύχη· ἡ γὰρ προαίρεσις οὐκ ἄνευ διανοίας.
thought and chance; the for preference is not without thought.

ἀόριστα μὲν οὖν τὰ αἴτια ἀνάγκη εἶναι ἀφ' ὧν ἂν
indefinitely on the one hand therefore the causes necessity to be from which ever
γένοιτο τὸ ἀπὸ τύχης. ὅθεν καὶ ἡ τύχη τοῦ ἀορίστου εἶναι δοκεῖ
may become the from luck. whence and the luck of the indefinite to be seems

καὶ ἄδηλος ἀνθρώπῳ, καὶ ἔστιν ὡς οὐδὲν ἀπὸ τύχης δόξειεν

and unclear to a human, and it is as nothing from luck it would seem

ἂν γίγνεσθαι. πάντα γὰρ ταῦτα ὀρθῶς λέγεται, εὐλόγως. ἔστιν

ever to happen. all for these correctly is said, reasonably. it is

μὲν γὰρ ὡς γίγνεται ἀπὸ τύχης· κατὰ συμβεβηκὸς γὰρ

on the one hand for as it happens from luck; according to coincidental for

γίγνεται, καὶ ἔστιν αἴτιον ὡς συμβεβηκὸς ἡ τύχη· ὡς δ' ἁπλῶς

it happens, and it is a cause as coincidental the luck; as but simply

οὐδενός· οἷον οἰκίας οἰκοδόμος μὲν αἴτιος, κατὰ

of nothing. such as a house builder on the one hand is responsible, whereas

συμβεβηκὸς δὲ αὐλητής,

coincidentally but a flutist,

καὶ τοῦ ἐλθόντα κομίσασθαι τὸ ἀργύριον, μὴ τούτου ἕνεκα

and of the one who comes to bring the money, not for this reason

ἐλθόντα, ἄπειρα τὸ πλῆθος· καὶ γὰρ ἰδεῖν τινὰ βουλόμενος καὶ

coming, infinite the multitude; and for to see some wishing and

διώκων καὶ φεύγων καὶ θεασόμενος. καὶ τὸ φάναι εἶναί τι

pursuing and fleeing and contemplating. and the to say to be something

παράλογον τὴν τύχην ὀρθῶς· ὁ γὰρ λόγος ἢ τῶν ἀεὶ ὄντων

irrational the fortune rightly; the for reason or of the always existing

ἢ τῶν ὡς ἐπὶ τὸ πολύ, ἡ δὲ τύχη ἐν τοῖς γιγνομένοις

or of the as concerning the many, the but fortune in the becoming

παρὰ ταῦτα. ὥστ' ἐπεὶ ἀόριστα τὰ οὕτως αἴτια, καὶ ἡ τύχη

contrary to these. So since indeterminate the such causes, and the fortune

ἀόριστον. ὅμως δ' ἐπ' ἐνίων ἀπορήσειεν ἄν τις, ἆρ' οὖν

is indeterminate. However and upon some would wonder if anyone, whether therefore

τὰ τυχόντα αἴτι' ἂν γένοιτο τῆς τύχης· οἷον ὑγιείας ἢ πνεῦμα

the occurrences cause would become of the fortune; such as health or spirit

ἢ εἵλησις, ἀλλ' οὐ τὸ ἀποκεκάρθαι· ἔστιν γὰρ ἄλλα ἄλλων
or selection, but not the to have been corrected; is for other of others
ἐγγύτερα τῶν κατὰ συμβεβηκὸς αἰτίων.
closer of the according to having happened causes.

τύχη δὲ ἀγαθὴ μὲν λέγεται ὅταν ἀγαθόν τι ἀποβῇ, φαύλη
luck but good indeed is said when good something happens, bad
δὲ ὅταν φαῦλόν τι, εὐτυχία δὲ καὶ δυστυχία ὅταν μέγεθος
but when bad something, good fortune but also misfortune when magnitude
ἔχοντα ταῦτα· διὸ καὶ τὸ παρὰ μικρὸν κακὸν ἢ ἀγαθὸν
having these things. therefore also the from small evil or good
λαβεῖν μέγα ἢ εὐτυχεῖν ἢ ἀτυχεῖν ἐστίν, ὅτι ὡς ὑπάρχον
to receive great or to be fortunate or to be unfortunate is, that as existing
λέγει ἡ διάνοια· τὸ γὰρ παρὰ μικρὸν ὥσπερ οὐδὲν ἀπέχειν
says the mind; the for from small just as nothing to be distant
δοκεῖ. ἔτι ἀβέβαιον ἡ εὐτυχία εὐλόγως· ἡ γὰρ τύχη ἀβέβαιος·
it seems. still uncertain the fortune reasonably; the for fortune is uncertain;
οὔτε γὰρ ἀεὶ οὔθ' ὡς ἐπὶ τὸ πολὺ οἷόν τ' εἶναι τῶν ἀπὸ τύχης
nor for always nor as in the much such and to be of those from fortune
οὐθέν.
nothing.

ἔστι μὲν οὖν ἄμφω αἴτια, καθάπερ εἴρηται, κατὰ
there is indeed therefore both causes, as it has been said, according to
συμβεβηκός-καὶ ἡ τύχη καὶ τὸ αὐτόματον-ἐν τοῖς ἐνδεχομένοις
that which has happened-and the chance and the automatic-in the possible
γίγνεσθαι μὴ ἁπλῶς μηδ' ὡς ἐπὶ τὸ πολύ, καὶ τούτων ὅσ'
things to occur not simply nor as regarding the many, and of these as many as

ἂν γένοιτο ἕνεκά του.

if it could happen for the sake of this.

Κεφάλαιο 6

Chapter 6

Διαφέρει δ' ὅτι τὸ αὐτόματον ἐπὶ πλεῖόν ἐστι· τὸ μὲν

It differs that because the automatic regarding more it is: the on one hand

γὰρ ἀπὸ τύχης πᾶν ἀπὸ ταὐτομάτου, τοῦτο δ' οὐ πᾶν

for from chance all from the automatic, this on the other hand not all

[197b] ἀπὸ τύχης.

[197b] from chance.

ἡ μὲν γὰρ τύχη καὶ τὸ ἀπὸ τύχης ἐστὶν ὅσοις καὶ τὸ

the on the one hand for fortune and the from fortune is to those and the

εὐτυχῆσαι ἂν ὑπάρξειεν καὶ ὅλως πρᾶξις. διὸ καὶ ἀνάγκη

to prosper if it were to exist and generally action. therefore and necessity

περὶ τὰ πρακτὰ εἶναι τὴν τύχην (σημεῖον δ' ὅτι δοκεῖ ἤτοι

concerning the actions to be the fortune (a sign that that seems either

ταὐτὸν εἶναι τῇ εὐδαιμονίᾳ ἡ εὐτυχία ἢ ἐγγύς, ἡ δ' εὐδαιμονία

the same to be to the happiness the good fortune or near, the and happiness

πρᾶξίς τις· εὐπραξία γάρ), ὥσθ' ὁπόσοις μὴ ἐνδέχεται πρᾶξαι,

action some; good action for), so that for those not able to act,

οὐδὲ τὸ ἀπὸ τύχης τι ποιῆσαι. καὶ διὰ τοῦτο οὔτε

nor the from fortune anything to achieve. and because of this neither

ἄψυχον οὐδὲν οὔτε θηρίον οὔτε παιδίον οὐδὲν ποιεῖ ἀπὸ τύχης,

lifeless anything nor beast nor child anything does from fortune,

ὅτι οὐκ ἔχει προαίρεσιν· οὐδ' εὐτυχία οὐδ' ἀτυχία ὑπάρχει

because not has volition; nor good fortune nor bad fortune exists

τούτοις, εἰ μὴ καθ' ὁμοιότητα, ὥσπερ ἔφη Πρώταρχος εὐτυχεῖς

these, if not according to similarity, just as said Protagoras fortunate

εἶναι τοὺς λίθους ἐξ ὧν οἱ βωμοί, ὅτι τιμῶνται, οἱ δὲ

to be the stones out of of which the altars, since they are honored, the and

ὁμόζυγες αὐτῶν καταπατοῦνται. τὸ δὲ πάσχειν ἀπὸ τύχης

those that are like them of them are trampled. the and suffering from fortune

ὑπάρξει πως καὶ τούτοις, ὅταν ὁ πράττων τι περὶ αὐτὰ

will exist somehow and these, when the one acting something regarding them

πράξῃ ἀπὸ τύχης, ἄλλως δὲ οὐκ ἔστιν·

does from fortune, otherwise and not it is,

τὸ δ' αὐτόματον καὶ τοῖς ἄλλοις ζώοις καὶ πολλοῖς τῶν ἀψύχων,

the and automatic and to the other animals and many of the inanimate,

οἷον ὁ ἵππος αὐτόματος, φαμέν, ἦλθεν, ὅτι ἐσώθη μὲν

such as the horse automatic, we say, came, because he was saved indeed

ἐλθών, οὐ τοῦ σωθῆναι δὲ ἕνεκα ἦλθε· καὶ ὁ τρίπους

having come, not to be saved but for the sake of he came. and the tripod

αὐτόματος κατέπεσεν· ἔστη μὲν γὰρ τοῦ καθῆσθαι ἕνεκα,

automatic fell down. it stood indeed for of the to sit for the sake of,

ἀλλ' οὐ τοῦ καθῆσθαι ἕνεκα κατέπεσεν.

but not of the to sit for the sake of he/she/it fell.

ὥστε φανερὸν ὅτι ἐν τοῖς ἁπλῶς ἕνεκά του γιγνομένοις, ὅταν

thus evident that in the simply for the sake of it happening, when

μὴ τοῦ συμβάντος ἕνεκα γένηται ὧν ἔξω τὸ αἴτιον, τότε

not of the event for the sake occurs of which outside the cause, then

ἀπὸ τοῦ αὐτομάτου λέγομεν· ἀπὸ τύχης δέ, τούτων ὅσα ἀπὸ τοῦ

from the automatic we say: from chance but, of these as many from the

αὐτομάτου γίγνεται τῶν προαιρετῶν τοῖς ἔχουσι προαίρεσιν.

automatic happens of the volitional to those having choice.

σημεῖον δὲ τὸ μάτην, ὅτι λέγεται ὅταν μὴ γένηται τὸ ἕνεκα

A sign but the vain, that it is said when not occurs the sake

ἄλλου ἐκείνου ἕνεκα, οἷον εἰ τὸ βαδίσαι λαπάξεως ἕνεκά

of another that for the sake, such as if the to walk of a fall, for the sake

ἐστιν, εἰ δὲ μὴ ἐγένετο βαδίσαντι, μάτην φαμὲν βαδίσαι

it is, if but not it happened to someone walking, in vain we say to walk

καὶ ἡ βάδισις ματαία, ὡς τοῦτο ὂν τὸ μάτην, τὸ πεφυκὸς

and the walking is in vain, as this being the vain, the natural

ἄλλου ἕνεκα, ὅταν μὴ περαίνῃ ἐκεῖνο οὗ ἕνεκα ἦν

of another sake, when not complete that of which for the sake of was

καὶ ἐπεφύκει, ἐπεὶ εἴ τις λούσασθαι φαίη μάτην ὅτι οὐκ

and had arisen, since if someone should wash would say in vain that not

ἐξέλιπεν ὁ ἥλιος, γελοῖος ἂν εἴη· οὐ γὰρ ἦν τοῦτο ἐκείνου

had failed the sun, ridiculous would be if it were; not for was this of that

ἕνεκα. οὕτω δὴ τὸ αὐτόματον καὶ κατὰ τὸ ὄνομα

for the sake of. Thus indeed the automatic and according to the name

ὅταν αὐτὸ μάτην γένηται· κατέπεσεν γὰρ οὐ τοῦ πατάξαι

whenever it in vain may come to be; had fallen for not of the to strike

ἕνεκεν ὁ λίθος· ἀπὸ τοῦ αὐτομάτου ἄρα κατέπεσεν ὁ λίθος,

for the sake of the stone; from the automatic thus had fallen the stone,

ὅτι πέσοι ἂν ὑπὸ τινὸς καὶ τοῦ πατάξαι ἕνεκα. μάλιστα

that it would fall if by someone and the to strike for the sake of. especially

δ' ἐστὶ χωριζόμενον τοῦ ἀπὸ τύχης ἐν τοῖς φύσει γιγνομένοις·

but it is separated from the from fate in the nature occurring;

ὅταν γὰρ γένηταί τι παρὰ φύσιν, τότε οὐκ ἀπὸ τύχης ἀλλὰ

whenever for it may happen something from nature, then not from chance but

μᾶλλον ἀπὸ ταὐτομάτου γεγονέναι φαμέν. ἔστι δὲ καὶ τοῦτο

rather from necessity to have come about we say. there is but also this

ἕτερον· τοῦ μὲν γὰρ ἔξω τὸ αἴτιον, τοῦ δ' ἐντός. [198a]

another; of the one for outside the cause, of the other inside. [198a]

τί μὲν οὖν ἐστιν τὸ αὐτόματον καὶ τί ἡ τύχη, εἴρηται,

what indeed therefore is the automatic and what the fortune, it has been said,

καὶ τί διαφέρουσιν ἀλλήλων. τῶν δὲ τρόπων τῆς αἰτίας ἐν

and what they differ from one another. of the but ways of the cause in

τοῖς ὅθεν ἡ ἀρχὴ τῆς κινήσεως ἑκάτερον αὐτῶν· ἢ γὰρ

the from where the beginning of the motion each of them; or for

τῶν φύσει τι ἢ τῶν ἀπὸ διανοίας αἰτίων ἀεί ἐστιν· ἀλλὰ

of the nature anything or of the from reason causes always is; but

τούτων τὸ πλῆθος ἀόριστον.

of these the multitude is indefinite.

ἐπεὶ δ' ἐστὶ τὸ αὐτόματον καὶ ἡ τύχη αἴτια ὧν ἂν ἢ νοῦς

since but is the automatic and the fortune cause of which whenever or mind

γένοιτο αἴτιος ἢ φύσις, ὅταν κατὰ συμβεβηκὸς αἴτιόν τι

becomes the cause or nature, when according to a chance event cause something

γένηται τούτων αὐτῶν, οὐδὲν δὲ κατὰ συμβεβηκός ἐστι

comes to be of these themselves, nothing but according to a chance event is

πρότερον τῶν καθ' αὑτό, δῆλον ὅτι οὐδὲ τὸ κατὰ

prior than the concerning itself, it is clear that nor the according to

συμβεβηκὸς αἴτιον πρότερον τοῦ καθ' αὑτό. ὕστερον ἄρα τὸ

accidental cause earlier of the by itself. later therefore the

αὐτόματον καὶ ἡ τύχη καὶ νοῦ καὶ φύσεως· ὥστ' εἰ ὅτι μάλιστα

automatic and the luck and mind and of nature. so that if that especially

τοῦ οὐρανοῦ αἴτιον τὸ αὐτόματον, ἀνάγκη πρότερον νοῦν αἴτιον

of the sky cause the automatic, necessity earlier mind cause

καὶ φύσιν εἶναι καὶ ἄλλων πολλῶν καὶ τοῦδε τοῦ παντός.

and nature to be and of others many and of this of the all.

Κεφάλαιο 7

Chapter 7

Ὅτι δὲ ἔστιν αἴτια, καὶ ὅτι τοσαῦτα τὸν ἀριθμὸν ὅσα φαμέν,

That but there is a cause, and that so many of the number as many we say,

δῆλον· τοσαῦτα γὰρ τὸν ἀριθμὸν τὸ διὰ τί περιείληφεν· ἢ

is evident. So many for the number the by which cause it is encompassed; or

γὰρ εἰς τὸ τί ἐστιν ἀνάγεται τὸ διὰ τί ἔσχατον, ἐν τοῖς

for into the cause there is derived, the by which cause ultimate, in the

ἀκινήτοις (οἷον ἐν τοῖς μαθήμασιν· εἰς ὁρισμὸν γὰρ τοῦ εὐθέος

immovable (such as in the studies; into definition for of the straight

ἢ συμμέτρου ἢ ἄλλου τινὸς ἀνάγεται ἔσχατον), ἢ εἰς τὸ

or of the proportional or of another something is derived ultimate), or into the

κινῆσαν πρῶτον (οἷον διὰ τί ἐπολέμησαν; ὅτι ἐσύλησαν), ἢ

moving first (such as by which cause they fought; that they plundered), or

τίνος ἕνεκα (ἵνα ἄρξωσιν), ἢ ἐν τοῖς γιγνομένοις ἡ ὕλη. ὅτι

of which reason (so that they may begin), or in the becoming the matter. That

μὲν οὖν τὰ αἴτια ταῦτα καὶ τοσαῦτα, φανερόν·

indeed therefore the causes these and so many, is clear.

ἐπεὶ δ' αἱ αἰτίαι τέτταρες, περὶ πασῶν τοῦ φυσικοῦ εἰδέναι, καὶ

since but the causes four, about all of the natural to know, and

εἰς πάσας ἀνάγων τὸ διὰ τί ἀποδώσει φυσικῶς, τὴν ὕλην, τὸ
for all necessities the through what will give back naturally, the matter, the
εἶδος, τὸ κινῆσαν, τὸ οὗ ἕνεκα. ἔρχεται δὲ τὰ τρία εἰς [τὸ] ἓν
form, the mover, the of which because. it comes but the three into [the] one
πολλάκις· τὸ μὲν γὰρ τί ἐστι καὶ τὸ οὗ ἕνεκα ἕν ἐστι, τὸ
many times; the indeed for what is and the of which because one is, the
δ' ὅθεν ἡ κίνησις πρῶτον τῷ εἴδει ταὐτὸ τούτοις· ἄνθρωπος
but whence the motion first to the form the same to these; man
γὰρ ἄνθρωπον γεννᾷ-καὶ ὅλως ὅσα κινούμενα κινεῖ (ὅσα
for man begets - and altogether all that move moves (all that
δὲ μή, οὐκέτι φυσικῆς· οὐ γὰρ ἐν αὐτοῖς ἔχοντα κίνησιν οὐδ'
but not, no longer of nature; not for in themselves having motion nor
ἀρχὴν κινήσεως κινεῖ, ἀλλ' ἀκίνητα ὄντα· διὸ τρεῖς αἱ
beginning of motion moves, but unmoving beings therefore there are three the
πραγματεῖαι, ἡ μὲν περὶ ἀκινήτων, ἡ δὲ περὶ κινουμένων μὲν
treatises, the one about unmoving things, the other about moving things one
ἀφθάρτων δέ, ἡ δὲ περὶ τὰ φθαρτά).
of the imperishable and, the other about the perishable things).

ὥστε τὸ διὰ τί καὶ εἰς τὴν ὕλην ἀνάγοντι ἀποδίδοται, καὶ εἰς τὸ
so the for what and into the matter leading is attributed, and into the
τί ἐστιν, καὶ εἰς τὸ πρῶτον κινῆσαν. περὶ γενέσεως γὰρ
what is, and into the first moved. concerning generation for
μάλιστα τοῦτον τὸν τρόπον τὰς αἰτίας σκοποῦσι, τί μετὰ τί
particularly this the way the causes they seek, what after what
γίγνεται, καὶ τί πρῶτον ἐποίησεν ἢ τί ἔπαθεν, καὶ οὕτως αἰεὶ τὸ
happens, and what first created or what suffered, and thus always the

ἐφεξῆς.

subsequent.

διτταὶ δὲ αἱ ἀρχαὶ αἱ κινοῦσαι φυσικῶς, ὧν ἡ ἑτέρα οὐ

two but the principles the moving naturally, of which the other not

φυσική· οὐ [198b] γὰρ ἔχει κινήσεως ἀρχὴν ἐν αὑτῇ. τοιοῦτον

natural; not [198b] for has of motion a principle in itself. such

δ' ἐστὶν εἴ τι κινεῖ μὴ κινούμενον, ὥσπερ τό τε παντελῶς

indeed is if anything moves not being moved, just as the and completely

ἀκίνητον καὶ [τὸ] πάντων πρῶτον καὶ τὸ τί ἐστιν καὶ ἡ μορφή·

unmoving and [the] of all first and the what is and the form;

τέλος γὰρ καὶ οὗ ἕνεκα· ὥστε ἐπεὶ ἡ φύσις ἕνεκά του, καὶ

for for and of which sake; so that since the nature sake of this, and

ταύτην εἰδέναι δεῖ, καὶ πάντως ἀποδοτέον τὸ διὰ τί, οἷον

this to know, is necessary, and certainly must be returned the for what, such as

ὅτι ἐκ τοῦδε ἀνάγκη τόδε (τὸ δὲ ἐκ τοῦδε ἢ ἁπλῶς ἢ ὡς ἐπὶ τὸ

that out of this necessity this (the but of this or simply or as on the

πολύ), καὶ εἰ μέλλει τοδὶ ἔσεσθαι (ὥσπερ ἐκ τῶν προτάσεων

many), and if it is about to this to be (just as out of the premises

τὸ συμπέρασμα), καὶ ὅτι τοῦτ' ἦν τὸ τί ἦν εἶναι, καὶ διότι βέλτιον

the conclusion), and that this was the what was to be, and because better

οὕτως, οὐχ ἁπλῶς, ἀλλὰ τὸ πρὸς τὴν ἑκάστου οὐσίαν.

thus, not simply, but the towards the each essence.

Κεφάλαιο 8

Chapter 8

Λεκτέον δὴ πρῶτον μὲν διότι ἡ φύσις τῶν ἕνεκά

It is to be said indeed first on the one hand because the nature of the for the sake of

του αἰτίων, ἔπειτα περὶ τοῦ ἀναγκαίου, πῶς ἔχει ἐν τοῖς

it causes, then regarding the necessary, how it exists in the

φυσικοῖς· εἰς γὰρ ταύτην τὴν αἰτίαν ἀνάγουσι πάντες, ὅτι ἐπειδὴ

natural. For for this the cause they lead all, because since

τὸ θερμὸν τοιονδὶ πέφυκεν καὶ τὸ ψυχρὸν καὶ ἕκαστον δὴ τῶν

the warm such has arisen and the cold and each indeed of the

τοιούτων, ταδὶ ἐξ ἀνάγκης ἐστὶ καὶ γίγνεται· καὶ γὰρ ἐὰν ἄλλην

such things, these from necessity it is and it becomes. And for if another

αἰτίαν εἴπωσιν, ὅσον ἁψάμενοι χαίρειν ἐῶσιν, ὁ μὲν τὴν

cause they say, as long as they are touched to rejoice, they let it be, the one the

φιλίαν καὶ τὸ νεῖκος, ὁ δὲ τὸν νοῦν·

friendship and the strife, the other the mind.

ἔχει δ' ἀπορίαν τί κωλύει τὴν φύσιν μὴ ἕνεκά του ποιεῖν

has but a difficulty what hinders the nature not for the sake of it to do

μηδ' ὅτι βέλτιον, ἀλλ' ὥσπερ ὑει ὁ Ζεὺς οὐχ ὅπως τὸν σῖτον

nor that better, but just as rains the Zeus not so that the grain

αὐξήσῃ, ἀλλ' ἐξ ἀνάγκης (τὸ γὰρ ἀναχθὲν ψυχθῆναι

he may increase, but out of necessity (the for having been raised to be cooled

δεῖ, καὶ τὸ ψυχθὲν ὕδωρ γενόμενον κατελθεῖν· τὸ δ'

must, and the having been cooled water having become to descend; the but

αὐξάνεσθαι τούτου γενομένου τὸν σῖτον συμβαίνει), ὁμοίως δὲ

to increase of this having happened the grain it happens), similarly and

καὶ εἴ τῳ ἀπόλλυται ὁ σῖτος ἐν τῇ ἅλῳ, οὐ τούτου ἕνεκα

also if to the is lost the grain in the salt, not of this for the sake of

ὑει ὅπως ἀπόληται, ἀλλὰ τοῦτο συμβέβηκεν-ὥστε τί κωλύει

it rains so that it may perish, but this has happened - so that what hinders

οὕτω καὶ τὰ μέρη ἔχειν ἐν τῇ φύσει, οἷον τοὺς ὀδόντας ἐξ

thus and the parts to have in the nature, such as the teeth out of

ἀνάγκης ἀνατεῖλαι τοὺς μὲν ἐμπροσθίους ὀξεῖς, ἐπιτηδείους

necessity to shine the on one hand forward sharp, suitable

πρὸς τὸ διαιρεῖν, τοὺς δὲ γομφίους πλατεῖς καὶ χρησίμους

for the dividing, the on the other hand molars wide and useful

πρὸς τὸ λεαίνειν τὴν τροφήν, ἐπεὶ οὐ τούτου ἕνεκα γενέσθαι,

for the grinding the food, since not of this reason to occur,

ἀλλὰ συμπεσεῖν· ὁμοίως δὲ καὶ περὶ τῶν ἄλλων μερῶν,

but to fall together. likewise also and concerning the other parts,

ἐν ὅσοις δοκεῖ ὑπάρχειν τὸ ἕνεκά του. ὅπου μὲν οὖν

in which seem to exist the reason of it. wherever on one hand therefore

ἅπαντα συνέβη ὥσπερ κἂν εἰ ἕνεκά του ἐγίγνετο, ταῦτα

all happened as even if if for the sake of it happened, these

μὲν ἐσώθη ἀπὸ τοῦ αὐτομάτου συστάντα ἐπιτηδείως· ὅσα

on one hand were saved from the automatic organization suitably. as many

δὲ μὴ οὕτως, ἀπώλετο καὶ ἀπόλλυται, καθάπερ Ἐμπεδοκλῆς

on the other hand not so, were lost and are being lost, just as Empedocles

λέγει τὰ βουγενῆ ἀνδρόπρωρα.

says the of beasts of human form.

ὁ μὲν οὖν λόγος, ᾧ ἄν τις ἀπορήσειεν, οὗτος,

the on the one hand therefore argument, to which anyone someone would wonder, this,

καὶ εἴ τις ἄλλος τοιοῦτός ἐστιν· ἀδύνατον δὲ τοῦτον ἔχειν τὸν

and if someone else such exists: impossible but this to have the

τρόπον. ταῦτα μὲν γὰρ καὶ πάντα τὰ φύσει ἢ αἰεὶ οὕτω

way. These on the one hand for and all the by nature or always thus

γίγνεται ἢ ὡς ἐπὶ τὸ πολύ, τῶν δ' ἀπὸ τύχης καὶ τοῦ

occurs or as in the case of the many, of those but from chance and the

αὐτομάτου οὐδέν. οὐ [199a] γὰρ ἀπὸ τύχης οὐδ' ἀπὸ συμπτώματος

automatic nothing. not [199a] for from chance nor from accident

δοκεῖ ὕειν πολλάκις τοῦ χειμῶνος, ἀλλ' ἐὰν ὑπὸ κύνα· οὐδὲ

seem to rain often of the winter, but if under a dog: nor

καύματα ὑπὸ κύνα, ἀλλ' ἂν χειμῶνος. εἰ οὖν ἢ ἀπὸ συμπτώματος

heats by a dog, but if of winter. if then or from accident

δοκεῖ ἢ ἕνεκά του εἶναι, εἰ μὴ οἷόν τε ταῦτ' εἶναι μήτε ἀπὸ

seems or for the sake of it to be, if not such and these to be nor from

συμπτώματος μήτ' ἀπὸ ταὐτομάτου, ἕνεκά του ἂν εἴη. ἀλλὰ

accident nor from spontaneity, on account of it if it were. but

μὴν φύσει γ' ἐστὶ τὰ τοιαῦτα πάντα, ὡς κἂν αὐτοὶ φαῖεν

indeed by nature indeed is the such all, as even if they should say

οἱ ταῦτα λέγοντες. ἔστιν ἄρα τὸ ἕνεκά του ἐν τοῖς φύσει

the these speaking. it is therefore the on account of it in the nature

γιγνομένοις καὶ οὖσιν.

beings and existing.

ἔτι ἐν ὅσοις τέλος ἔστι τι, τούτου ἕνεκα πράττεται τὸ

still in as many as end is something, of this sake is done the

πρότερον καὶ τὸ ἐφεξῆς. οὐκοῦν ὡς πράττεται, οὕτω πέφυκε, καὶ

former and the latter. therefore as is done, thus has become, and

ὡς πέφυκεν, οὕτω πράττεται ἕκαστον, ἂν μή τι ἐμποδίζῃ.

as it has come to be, thus is done each, if not something hinders.

πράττεται δ' ἕνεκά του· καὶ πέφυκεν ἄρα ἕνεκά του.

is done but for the sake of this. and has become therefore for the sake of this.

οἷον εἰ οἰκία τῶν φύσει γιγνομένων ἦν, οὕτως ἂν ἐγίγνετο
such as if house of those by nature coming to be was, thus would come to be
ὡς νῦν ὑπὸ τῆς τέχνης· εἰ δὲ τὰ φύσει μὴ μόνον φύσει ἀλλὰ
as now by the art; if but the by nature not only by nature but also
καὶ τέχνῃ γίγνοιτο, ὡσαύτως ἂν γίγνοιτο ᾗ πέφυκεν.
and by art would come to be, similarly would come to be in which it has come to be.
ἕνεκα ἄρα θατέρου θάτερον. ὅλως δὲ ἡ τέχνη τὰ μὲν
for the sake therefore of the other one. altogether but the art the indeed
ἐπιτελεῖ ἃ ἡ φύσις ἀδυνατεῖ ἀπεργάσασθαι, τὰ δὲ μιμεῖται. εἰ
accomplishes what the nature is unable to accomplish, the but imitates. if
οὖν τὰ κατὰ τέχνην ἕνεκά του, δῆλον ὅτι καὶ τὰ
therefore the according to art for the sake of it, it is clear that also the
κατὰ φύσιν· ὁμοίως γὰρ ἔχει πρὸς ἄλληλα ἐν τοῖς κατὰ
according to nature; similarly for it has towards each other in the according to
τέχνην καὶ ἐν τοῖς κατὰ φύσιν τὰ ὕστερα πρὸς τὰ πρότερα.
art and in the according to nature the later towards the earlier.
μάλιστα δὲ φανερὸν ἐπὶ τῶν ζῴων τῶν ἄλλων, ἃ οὔτε τέχνῃ
especially but evident in the animals of the others, which neither by art
οὔτε ζητήσαντα οὔτε βουλευσάμενα ποιεῖ· ὅθεν διαποροῦσί τινες
nor having sought nor having planned does. whence they are perplexed some
πότερον νῷ ἢ τινι ἄλλῳ ἐργάζονται οἵ τ’ ἀράχναι καὶ
whether by thought or by some other are working those and spiders and
οἱ μύρμηκες καὶ τὰ τοιαῦτα. κατὰ μικρὸν δ’ οὕτω προϊόντι
the ants and the such things. according to small but thus advancing
καὶ ἐν τοῖς φυτοῖς φαίνεται τὰ συμφέροντα γιγνόμενα πρὸς τὸ
and in the plants appears the interests arising towards the
τέλος, οἷον τὰ φύλλα τῆς τοῦ καρποῦ ἕνεκα σκέπης.
end, such as the leaves of the the fruit for the sake of shelter.

ὥστ' εἰ φύσει τε ποιεῖ καὶ ἕνεκά του ἡ χελιδὼν τὴν
so that if by nature and makes and for the sake of it the swallow the
νεοττιὰν καὶ ὁ ἀράχνης τὸ ἀράχνιον, καὶ τὰ φυτὰ τὰ φύλλα
nest and the spider the web, and the plants the leaves
ἕνεκα τῶν καρπῶν καὶ τὰς ῥίζας οὐκ ἄνω ἀλλὰ κάτω τῆς
for the sake of the fruits and the roots not above but below the
τροφῆς, φανερὸν ὅτι ἔστιν ἡ αἰτία ἡ τοιαύτη ἐν τοῖς φύσει
nourishment, evident that there is the cause the such in the nature
γιγνομένοις καὶ οὖσιν. καὶ ἐπεὶ ἡ φύσις διττή, ἡ μὲν ὡς ὕλη
arising and existing. And since the nature is dual, the one as matter
ἡ δ' ὡς μορφή, τέλος δ' αὕτη, τοῦ τέλους δὲ ἕνεκα
the other as form, end but this, of the end but for the sake of
τἄλλα, αὕτη ἂν εἴη ἡ αἰτία, ἡ οὗ ἕνεκα. ἁμαρτία δὲ
the others, this would be the cause, the of which for the sake of. Sin but
γίγνεται καὶ ἐν τοῖς κατὰ τέχνην (ἔγραψε γὰρ οὐκ ὀρθῶς ὁ
occurs and in those according to art (he wrote for not correctly the
γραμματικός, καὶ ἐπότισεν [οὐκ ὀρθῶς] ὁ ἰατρὸς τὸ φάρμακον),
grammarian, and he watered [not correctly] the physician the medicine),
ὥστε δῆλον ὅτι ἐνδέχεται καὶ ἐν τοῖς κατὰ [199b] φύσιν.
so that it is clear that it is possible also in those according to [199b] nature.
εἰ δὴ ἔστιν ἕνια κατὰ τέχνην ἐν οἷς τὸ ὀρθῶς ἕνεκά
if indeed there is some according to art in which the correctly for the sake of
του, ἐν δὲ τοῖς ἁμαρτανομένοις ἕνεκα μέν τινος ἐπιχειρεῖται
it, in but those erring for the sake of indeed some attempts
ἀλλ' ἀποτυγχάνεται, ὁμοίως ἂν ἔχοι καὶ ἐν τοῖς φυσικοῖς, καὶ
but fails, likewise would have and in those natural, and
τὰ τέρατα ἁμαρτήματα ἐκείνου τοῦ ἕνεκά του. καὶ ἐν ταῖς
the monsters sins of that the for the sake of it. and in the

ἐξ ἀρχῆς ἄρα συστάσεσι τὰ βουγενῆ, εἰ μὴ πρός τινα ὅρον καὶ
from beginning thus compositions the of the will, if not towards some limit and
τέλος δυνατὰ ἦν ἐλθεῖν, διαφθειρομένης ἂν ἀρχῆς τινὸς
end possible it was to come, being destroyed if of a beginning of some
ἐγίγνετο, ὥσπερ νῦν τοῦ σπέρματος.
it came into being, just as now of the seed.

ἔτι ἀνάγκη σπέρμα γενέσθαι πρῶτον, ἀλλὰ μὴ εὐθὺς τὰ ζῷα·
yet necessity seed to be born first, but not immediately the animals.
καὶ τὸ "οὐλοφυὲς μὲν πρῶτα" σπέρμα ἦν.
and the "evil indeed first" seed was.

ἔτι καὶ ἐν τοῖς φυτοῖς ἔνεστι τὸ ἕνεκά του, ἧττον δὲ διήρθρωται·
still and in the plants is the reason of it, less however it is altered.
πότερον οὖν καὶ ἐν τοῖς φυτοῖς ἐγίγνετο, ὥσπερ τὰ βουγενῆ
either therefore and in the plants it became, just as the bovine
ἀνδρόπρῳρα, οὕτω καὶ ἀμπελογενῆ ἐλαιόπρῳρα, ἢ οὔ; ἄτοπον γάρ·
anthropoids, so and grape-producing olive varieties, or not? strange for
ἀλλὰ μὴν ἔδει γε, εἴπερ καὶ ἐν τοῖς ζῴοις.
but indeed it was necessary certainly, if also in the animals.

ἔτι ἔδει καὶ ἐν τοῖς σπέρμασι γίγνεσθαι ὅπως ἔτυχεν·
still it was necessary also in the seeds to be formed in order that it happened;
ὅλως δ' ἀναιρεῖ ὁ οὕτως λέγων τὰ φύσει τε καὶ φύσιν· φύσει
entirely but it denies the such saying the by nature and and nature; by nature
γάρ, ὅσα ἀπό τινος ἐν αὑτοῖς ἀρχῆς συνεχῶς κινούμενα
for as much as from of some in themselves of the beginning continuously being moved
ἀφικνεῖται εἴς τι τέλος· ἀφ' ἑκάστης δὲ οὐ τὸ αὐτὸ ἑκάστοις
arrives at some end. from each and not the same to each

οὐδὲ τὸ τυχόν, ἀεὶ μέντοι ἐπὶ τὸ αὐτό, ἂν μή τι ἐμποδίσῃ.

nor the happening, always however upon the same, if not something hinders.

τὸ δὲ οὗ ἕνεκα, καὶ ὃ τούτου ἕνεκα, γένοιτο

the but of which for the sake of, and that which of this for the sake of, may it happen

ἂν καὶ ἀπὸ τύχης, οἷον λέγομεν ὅτι ἀπὸ τύχης ἦλθεν ὁ ξένος

if and from chance, such as we say that from chance it has come the stranger

καὶ λυ σάμενος ἀπῆλθεν, ὅταν ὥσπερ ἕνεκα τούτου

and having loosened having taken he departed, when just as for the sake of this

ἐλθὼν πράξῃ, μὴ ἕνεκα δὲ τούτου ἔλθῃ. καὶ τοῦτο

coming into action, not for the sake of but this may he come. and this

κατὰ συμβεβηκός (ἡ γὰρ τύχη τῶν κατὰ συμβεβηκὸς

according to what happens (the for chance of those according to accidental

αἰτίων, καθάπερ καὶ πρότερον εἴπομεν), ἀλλ' ὅταν τοῦτο αἰεὶ ἢ

causes, just as and previously we have said), but whenever this always or

ὡς ἐπὶ τὸ πολὺ γένηται, οὐ συμβεβηκὸς οὐδ' ἀπὸ τύχης· ἐν δὲ

as much the majority happens, not accidental nor from chance. in but

τοῖς φυσι κοῖς ἀεὶ οὕτως, ἂν μή τι ἐμποδίσῃ.

the natural things, always thus, if not something prevents.

ἄτοπον δὲ τὸ μὴ οἴεσθαι ἕνεκά του γίγνεσθαι, ἐὰν

unreasonable but the not to suppose for the sake of of this to become, if

μὴ ἴδωσι τὸ κινοῦν βουλευσάμενον. καίτοι καὶ ἡ τέχνη οὐ

not they see the moving having deliberated. And yet also the art does not

βουλεύεται· καὶ εἰ ἐνῆν ἐν τῷ ξύλῳ ἡ ναυπηγική, ὁμοίως ἂν

deliberate; and if there was in the wood, the shipbuilding, similarly would

τῇ φύσει ἐποίει· ὥστ' εἰ ἐν τῇ τέχνῃ ἔνεστι τὸ ἕνεκά

to the nature do; so that if in the art there is the for the sake of

του, καὶ ἐν τῇ φύσει. μάλιστα δὲ δῆλον, ὅταν τις ἰατρεύῃ
this, and in the nature. Most but it is clear, when someone treats
αὐτὸς ἑαυτόν· τούτῳ γὰρ ἔοικεν ἡ φύσις. ὅτι μὲν οὖν
himself; himself; to this for it is similar the nature. that on one hand therefore
αἰτία ἡ φύσις, καὶ οὕτως ὡς ἕνεκά του, φανερόν.
the cause the nature, and thus as for the sake of this, is evident.

Κεφάλαιο 9

Chapter 9

Τὸ δ' ἐξ ἀνάγκης πότερον ἐξ ὑποθέσεως ὑπάρχει ἢ καὶ ἁπλῶς;
The and from necessity whether from hypothesis exists or and simply?
νῦν μὲν γὰρ οἴονται τὸ ἐξ ἀνάγκης εἶναι [200a] ἐν τῇ
Now on the one hand for they think that from necessity to be [200a] in the
γενέσει ὥσπερ ἂν εἴ τις τὸν τοῖχον ἐξ ἀνάγκης γεγενῆσθαι
generation just as if anyone should the wall from necessity to have been generated
νομίζοι, ὅτι τὰ μὲν βαρέα κάτω πέφυκε φέρεσθαι
they would think, that the on the one hand heavy down has been borne to be carried
τὰ δὲ κοῦφα ἐπιπολῆς, διὸ οἱ λίθοι μὲν κάτω καὶ τὰ
the but light to the surface, therefore the stones on the one hand down and the
θεμέλια, ἡ δὲ γῆ ἄνω διὰ κουφότητα, ἐπιπολῆς δὲ μάλιστα
foundations, the but earth upwards through lightness, surface but especially
τὰ ξύλα· κουφότατα γάρ. ἀλλ' ὅμως οὐκ ἄνευ μὲν
the wood; the lightest indeed. but nevertheless, not without on the one hand
τούτων γέγονεν, οὐ μέντοι διὰ ταῦτα πλὴν ὡς δι' ὕλην,
these it has come to be, not however because of these except as through matter,
ἀλλ' ἕνεκα τοῦ κρύπτειν ἄττα καὶ σώζειν. ὁμοίως δὲ καὶ ἐν
but for the sake of the hiding anything and saving. Similarly but and in

τοῖς ἄλλοις πᾶσιν, ἐν ὅσοις τὸ ἕνεκά του ἔστιν, οὐκ ἄνευ μὲν
the others all, in as many as the sake of it exists, not without indeed
τῶν ἀναγκαίαν ἐχόντων τὴν φύσιν, οὐ μέντοι γε διὰ ταῦτα
the necessary having the nature, not however at least through these
ἀλλ' ἢ ὡς ὕλην, ἀλλ' ἕνεκά του, οἷον διὰ τί ὁ πρίων
but or as matter, but for the sake of it, such as through what the saw
τοιοσδί; ὅπως τοδὶ καὶ ἕνεκα τουδί. τοῦτο μέντοι τὸ οὗ
such? so that this and for the sake of this. This however the of which
ἕνεκα ἀδύνατον γενέσθαι, ἂν μὴ σιδηροῦς ᾖ· ἀνάγκη
for the sake of impossible to be, if not made of iron exists; necessity
ἄρα σιδηροῦν εἶναι, εἰ πρίων ἔσται καὶ τὸ ἔργον αὐτοῦ. ἐξ
therefore to be iron to be, if a saw will be and the work of it. From
ὑποθέσεως δὴ τὸ ἀναγκαῖον, ἀλλ' οὐχ ὡς τέλος· ἐν γὰρ τῇ ὕλῃ
assumption indeed the necessary, but not as end in for the matter
τὸ ἀναγκαῖον, τὸ δ' οὗ ἕνεκα ἐν τῷ λόγῳ.
the necessity, the but of which for the sake in the word.

ἔστι δὲ τὸ ἀναγκαῖον ἔν τε τοῖς μαθήμασι καὶ ἐν τοῖς κατὰ
is but the necessary in and the lessons and in the according to
φύσιν γιγνομένοις τρόπον τινὰ παραπλησίως· ἐπεὶ γὰρ τὸ εὐθὺ
nature things becoming manner some similarly. since for the straight
τοδί ἐστιν, ἀνάγκη τὸ τρίγωνον δύο ὀρθαῖς ἴσας ἔχειν· ἀλλ'
this is, necessity the triangle two right equal to have. but
οὐκ ἐπεὶ τοῦτο, ἐκεῖνο· ἀλλ' εἴ γε τοῦτο μὴ ἔστιν, οὐδὲ τὸ
not since this, that. but if indeed this not is, neither the
εὐθὺ ἔστιν. ἐν δὲ τοῖς γιγνομένοις ἕνεκά του ἀνάπαλιν, εἰ
straight is. in but the things becoming for the sake of that again, if

τὸ τέλος ἔσται ἢ ἔστι, καὶ τὸ ἔμπροσθεν ἔσται ἢ ἔστιν· εἰ
the end it will be or is, and the beforehand it will be or is. if
δὲ μή, ὥσπερ ἐκεῖ μὴ ὄντος τοῦ συμπεράσματος ἡ ἀρχὴ οὐκ
but not, just as there not being of the conclusion the beginning will not
ἔσται, καὶ ἐνταῦθα τὸ τέλος καὶ τὸ οὗ ἕνεκα. ἀρχὴ
be, and here the end and the of which for the sake of. beginning
γὰρ καὶ αὕτη, οὐ τῆς πράξεως ἀλλὰ τοῦ λογισμοῦ (ἐκεῖ δὲ τοῦ
for and this, not of the action but of the reasoning (there and of the
λογισμοῦ· πράξεις γὰρ οὐκ εἰσίν). ὥστ' εἰ ἔσται οἰκία,
reasoning· actions for not they are). so that if there will be a house,
ἀνάγκη ταῦτα γενέσθαι ἢ ὑπάρχειν, ἢ εἶναι [ἢ] ὅλως τὴν ὕλην
necessity these to happen or to exist, or to be [or] in total the matter
τὴν ἕνεκά του, οἷον πλίνθους καὶ λίθους, εἰ οἰκία·
the for the sake of it, such as bricks and stones, if there is a house·
οὐ μέντοι διὰ ταῦτά ἐστι τὸ τέλος ἀλλ' ἢ ὡς ὕλην, οὐδ'
not however through these things there is the end but or as matter, nor
ἔσται διὰ ταῦτα. ὅλως μέντοι μὴ ὄντων οὐκ ἔσται
will there be through these. altogether however not being not will there be
οὔθ' ἡ οἰκία οὔθ' ὁ πρίων, ἡ μὲν εἰ μὴ οἱ λίθοι, ὁ δ' εἰ
neither the house nor the saw, the indeed if not the stones, the but if
μὴ ὁ σίδηρος· οὐδὲ γὰρ ἐκεῖ αἱ ἀρχαί, εἰ μὴ τὸ τρίγωνον δύο
not the iron nor for there the beginnings, if not the triangle two
ὀρθαί.
right.

φανερὸν δὴ ὅτι τὸ ἀναγκαῖον ἐν τοῖς φυσικοῖς τὸ ὡς ὕλη λεγόμενον
evident indeed that the necessary in the natural the as matter called

καὶ αἱ κινήσεις αἱ ταύτης. καὶ ἄμφω μὲν τῷ φυσικῷ λεκτέαι

and the motions the of this. And both indeed the natural should be spoken

αἱ αἰτίαι, μᾶλλον δὲ ἡ τίνος ἕνεκα· αἴτιον γὰρ τοῦτο

the causes, rather but the of what for the sake of; cause for this

τῆς ὕλης, ἀλλ' οὐχ αὕτη τοῦ τέλους· καὶ τὸ τέλος τὸ οὗ

of the matter, but not this of the end; and the end the of which

ἕνεκα, καὶ ἡ ἀρχὴ ἀπὸ τοῦ ὁρισμοῦ καὶ τοῦ λόγου, ὥσπερ

for the sake, and the beginning from the definition and the discourse, just as

ἐν τοῖς κατὰ [200b] τέχνην, ἐπεὶ ἡ οἰκία τοιόνδε, τάδε δεῖ

in the according to [200b] art, since the house of this kind, these must

γενέσθαι καὶ ὑπάρχειν ἐξ ἀνάγκης, καὶ ἐπεὶ ἡ ὑγίεια τοδί, τάδε

come to be and exist out of necessity, and since the health of this, these

δεῖ γενέσθαι ἐξ ἀνάγκης καὶ ὑπάρχειν-οὕτως καὶ εἰ ἄνθρωπος

must come to be out of necessity and to exist thus and if a person

τοδί, ταδί· εἰ δὲ ταδί, ταδί. ἴσως δὲ καὶ ἐν τῷ λόγῳ ἔστιν τὸ

this, that· if but that, that. perhaps but also in the word it is the

ἀναγκαῖον. ὁρισαμένῳ γὰρ τὸ ἔργον τοῦ πρίειν ὅτι διαίρεσις

necessity. having defined for the work of the to prune that division

τοιαδί, αὕτη γ' οὐκ ἔσται, εἰ μὴ ἕξει ὀδόντας τοιουσδί· οὗτοι

such, this indeed not will be, if not it has teeth of such· these

δ' οὔ, εἰ μὴ σιδηροῦς. ἔστι γὰρ καὶ ἐν τῷ λόγῳ ἔνια μόρια ὡς ὕλη

but not, if not iron. it is for also in the word some parts as matter

τοῦ λόγου.

of the word.